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ANTICOAGULANT
QUICK GUIDE

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This publication is a compilation of selected important information on anticoagulants. The reader is encouraged to refer to the quoted references for further reading.

FOREWORD



The Anticoagulant Quick Guide is put together by the Clinical Pharmacy Working Committee (Cardiology Pharmacy Subspecialty) of the Pharmaceutical Services Programme, Ministry of Health Malaysia. It is a compilation of tables and charts from various latest references with that hope that it may serve to guide pharmacists in clinical practice, especially in the areas of Anticoagulation and Cardiology Pharmacy.

The issues covered by this publication are commonly encountered which include among others, dosing adjustments of anticoagulants, conversion of warfarin to direct oral anticoagulants (DOACs) and vice versa as well as the use of anticoagulants in special populations. The management of bleeding and reversal strategies of anticoagulants are also included as a good grasp of this knowledge is important to ensure patient safety.

Indeed, I would like to congratulate the committee for their contribution and commitment in producing this publication. May pharmacists in various settings find this compilation useful as a quick source of information and a gateway for more detailed explanation from the quoted references, if needed. May we continue to strive to work towards better patient care and safety.

Thank you.

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Warfarin

Warfarin Dose Adjustment

Indication :
Treatment and prophylaxis of thromboembolic disorders[1]

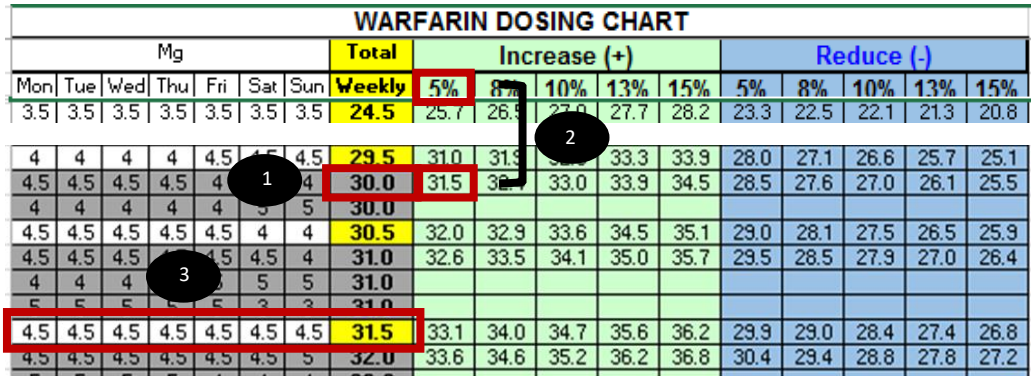
Increase dose:
According to ICSI[2] guideline, an increase of 15% weekly dose would correspond to an increase in International Normalized Ratio (INR) of 1. **Locally, for simplicity, we use 1% increase in warfarin dose will correspond to an increase in INR of 0.1**

Example of how to use the Warfarin Dosing Chart?

A **INCREASE DOSE**

1) For a patient with weekly dose of **30mg**, if you want to make a 5% increase in weekly dose
2) Look at the column for positive 5%, which is calculated to be 31.5mg
3) Going down the table for 31.5mg, you could see the dosage to be as follows:-

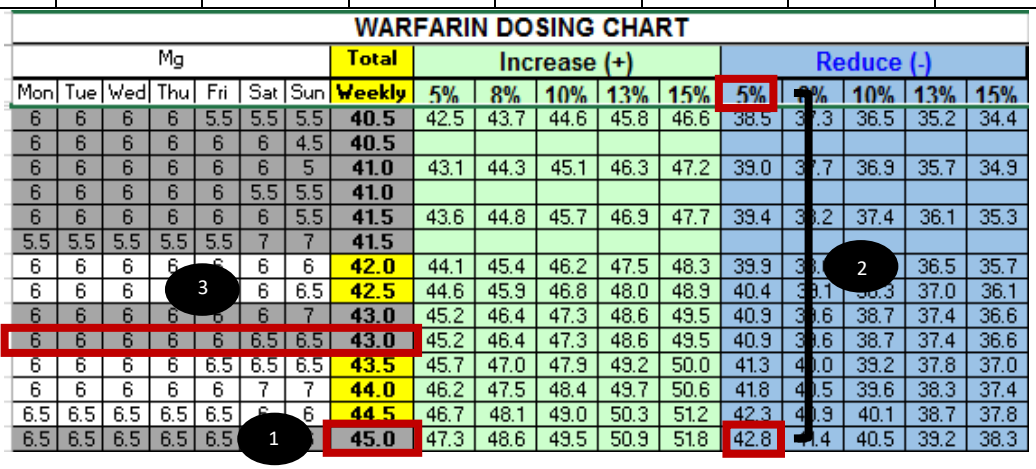
4.5	4.5	4.5	4.5	4.5	4.5	4.5	**31.5**



B **REDUCE DOSE**

1) For a patient with weekly dose of **45mg**, if you want to reduce the weekly dose by 5%
2) Look at the column for negative 5%, which is calculated to be 42.8mg
3) Going down the table for the nearest rounded up dose of 43mg, you could see the dosage to be as follows: -

6	6	6	6	6	6.5	6.5	**43**



Decrease dose: By stopping 1 day of warfarin, INR could reduce by an average of 0.26 (range from 0.2 - 0.5).
Dose should be reduced if the INR is persistently high. Please assume 0.3 – 0.5 reduction of INR if you are withholding 1 day of warfarin or assume that a 1% reduction in warfarin dose will correspond to a reduction in INR of 0.1

WARFARIN DOSING CHART

Mg							Total	Increase (+)					Reduce (-)				
Mon	Tue	Wed	Thu	Fri	Sat	Sun	Weekly	5%	8%	10%	13%	15%	5%	8%	10%	13%	15%
0.5	0.5	0.5	0.5	0.5	0.5	0	**3.0**	3.2	3.2	3.3	3.4	3.5	2.9	2.8	2.7	2.6	2.6
0.5	0.5	0.5	0.5	0.5	0.5	0.5	**3.5**	3.7	3.8	3.9	4.0	4.0	3.3	3.2	3.2	3.0	3.0
0.5	0.5	0.5	0.5	0.5	0.5	1	**4.0**	4.2	4.3	4.4	4.5	4.6	3.8	3.7	3.6	3.5	3.4
0.5	0.5	0.5	0.5	0.5	1	1	**4.5**	4.7	4.9	5.0	5.1	5.2	4.3	4.1	4.1	3.9	3.8
0.5	0.5	0.5	0.5	1	1	1	**5.0**	5.3	5.4	5.5	5.7	5.8	4.8	4.6	4.5	4.4	4.3
0.5	0.5	0.5	1	1	1	1	**5.5**	5.8	5.9	6.1	6.2	6.3	5.2	5.1	5.0	4.8	4.7
1	1	1	1	1	0.5	0.5	**6.0**	6.3	6.5	6.6	6.8	6.9	5.7	5.5	5.4	5.2	5.1
1	1	1	1	1	1	0.5	**6.5**	6.8	7.0	7.2	7.3	7.5	6.2	6.0	5.9	5.7	5.5
1	1	1	1	1	1	1	**7.0**	7.4	7.6	7.7	7.9	8.1	6.7	6.4	6.3	6.1	6.0
1	1	1	1	1	1	1.5	**7.5**	7.9	8.1	8.3	8.5	8.6	7.1	6.9	6.8	6.5	6.4
1	1	1	1	1	1.5	1.5	**8.0**	8.4	8.6	8.8	9.0	9.2	7.6	7.4	7.2	7.0	6.8
1	1	1	1	1.5	1.5	1.5	**8.5**	8.9	9.2	9.4	9.6	9.8	8.1	7.8	7.7	7.4	7.2
1.5	1.5	1.5	1.5	1	1	1	**9.0**	9.5	9.7	9.9	10.2	10.4	8.6	8.3	8.1	7.8	7.7
1.5	1.5	1.5	1.5	1.5	1	1	**9.5**	10.0	10.3	10.5	10.7	10.9	9.0	8.7	8.6	8.3	8.1
1.5	1.5	1.5	1.5	1.5	1.5	1	**10.0**	10.5	10.8	11.0	11.3	11.5	9.5	9.2	9.0	8.7	8.5
1.5	1.5	1.5	1.5	1.5	1.5	1.5	**10.5**	11.0	11.3	11.6	11.9	12.1	10.0	9.7	9.5	9.1	8.9
1.5	1.5	1.5	1.5	1.5	1.5	2	**11.0**	11.6	11.9	12.1	12.4	12.7	10.5	10.1	9.9	9.6	9.4
1.5	1.5	1.5	1.5	1.5	2	2	**11.5**	12.1	12.4	12.7	13.0	13.2	10.9	10.6	10.4	10.0	9.8
1.5	1.5	1.5	1.5	2	2	2	**12.0**	12.6	13.0	13.2	13.6	13.8	11.4	11.0	10.8	10.4	10.2
2	2	2	2	1.5	1.5	1.5	**12.5**	13.1	13.5	13.8	14.1	14.4	11.9	11.5	11.3	10.9	10.6
2	2	2	2	2	1.5	1.5	**13.0**	13.7	14.0	14.3	14.7	15.0	12.4	12.0	11.7	11.3	11.1
2	2	2	2	2	2	1.5	**13.5**	14.2	14.6	14.9	15.3	15.5	12.8	12.4	12.2	11.7	11.5
2	2	2	2	2	2	2	**14.0**	14.7	15.1	15.4	15.8	16.1	13.3	12.9	12.6	12.2	11.9
2	2	2	2	2	2	2.5	**14.5**	15.2	15.7	16.0	16.4	16.7	13.8	13.3	13.1	12.6	12.3
2	2	2	2	2	2.5	2.5	**15.0**	15.8	16.2	16.5	17.0	17.3	14.3	13.8	13.5	13.1	12.8
2	2	2	2	2.5	2.5	2.5	**15.5**	16.3	16.7	17.1	17.5	17.8	14.7	14.3	14.0	13.5	13.2
2	2	2	2	2	3	3	**16.0**	16.8	17.3	17.6	18.1	18.4	15.2	14.7	14.4	13.9	13.6
2.5	2.5	2.5	2.5	2.5	2	2	**16.5**	17.3	17.8	18.2	18.6	19.0	15.7	15.2	14.9	14.4	14.0
2.5	2.5	2.5	2.5	2.5	2.5	2	**17.0**	17.9	18.4	18.7	19.2	19.6	16.2	15.6	15.3	14.8	14.5
2.5	2.5	2.5	2.5	2.5	2.5	2.5	**17.5**	18.4	18.9	19.3	19.8	20.1	16.6	16.1	15.8	15.2	14.9
2.5	2.5	2.5	2.5	2.5	2.5	3	**18.0**	18.9	19.4	19.8	20.3	20.7	17.1	16.6	16.2	15.7	15.3
2.5	2.5	2.5	2.5	2.5	3	3	**18.5**	19.4	20.0	20.4	20.9	21.3	17.6	17.0	16.7	16.1	15.7
2.5	2.5	2.5	2.5	3	3	3	**19.0**	20.0	20.5	20.9	21.5	21.9	18.1	17.5	17.1	16.5	16.2
3	3	3	3	2.5	2.5	2.5	**19.5**	20.5	21.1	21.5	22.0	22.4	18.5	17.9	17.6	17.0	16.6
3	3	3	3	3	2.5	2.5	**20.0**	21.0	21.6	22.0	22.6	23.0	19.0	18.4	18.0	17.4	17.0
3	3	3	3	3	3	2	**20.0**										
3	3	3	3	3	3	2.5	**20.5**	21.5	22.1	22.6	23.2	23.6	19.5	18.9	18.5	17.8	17.4
3	3	3	3	3	3	3	**21.0**	22.1	22.7	23.1	23.7	24.2	20.0	19.3	18.9	18.3	17.9
3	3	3	3	3	3	3.5	**21.5**	22.6	23.2	23.7	24.3	24.7	20.4	19.8	19.4	18.7	18.3
3	3	3	3	3	3	4	**22.0**	23.1	23.8	24.2	24.9	25.3	20.9	20.2	19.8	19.1	18.7
3	3	3	3	3	3.5	3.5	**22.0**										
3	3	3	3	3.5	3.5	3.5	**22.5**	23.6	24.3	24.8	25.4	25.9	21.4	20.7	20.3	19.6	19.1
3	3	3	3	3	4	4	**23.0**	24.2	24.8	25.3	26.0	26.5	21.9	21.2	20.7	20.0	19.6
3.5	3.5	3.5	3.5	3.5	3	3	**23.5**	24.7	25.4	25.9	26.6	27.0	22.3	21.6	21.2	20.4	20.0
3.5	3.5	3.5	3.5	3.5	3.5	3	**24.0**	25.2	25.9	26.4	27.1	27.6	22.8	22.1	21.6	20.9	20.4
4	4	4	4	4	2	2	**24.0**										
3	3	3	3	3	4.5	4.5	**24.0**										
4	4	4	3	3	3	3	**24.0**										

WARFARIN DOSING CHART

Mg							Total	Increase (+)					Reduce (-)				
Mon	Tue	Wed	Thu	Fri	Sat	Sun	Weekly	5%	8%	10%	13%	15%	5%	8%	10%	13%	15%
3.5	3.5	3.5	3.5	3.5	3.5	3.5	**24.5**	25.7	26.5	27.0	27.7	28.2	23.3	22.5	22.1	21.3	20.8
4	4	4	4	4	2.5	2.5	**25.0**	26.3	27.0	27.5	28.3	28.8	23.8	23.0	22.5	21.8	21.3
3.5	3.5	3.5	3.5	3.5	3.5	4	**25.0**										
4	4	4	4	3	3	3	**25.0**										
3.5	3.5	3.5	3.5	3.5	4	4	**25.5**	26.8	27.5	28.1	28.8	29.3	24.2	23.5	23.0	22.2	21.7
3.5	3.5	3.5	3.5	4	4	4	**26.0**	27.3	28.1	28.6	29.4	29.9	24.7	23.9	23.4	22.6	22.1
4	4	4	4	4	3	3	**26.0**										
4	4	4	4	3.5	3.5	3.5	**26.5**	27.8	28.6	29.2	29.9	30.5	25.2	24.4	23.9	23.1	22.5
4	4	4	4	4	4	3	**27.0**	28.4	29.2	29.7	30.5	31.1	25.7	24.8	24.3	23.5	23.0
4	4	4	4	4	4	3.5	**27.5**	28.9	29.7	30.3	31.1	31.6	26.1	25.3	24.8	23.9	23.4
4	4	4	4	4	4	4	**28.0**	29.4	30.2	30.8	31.6	32.2	26.6	25.8	25.2	24.4	23.8
4	4	4	4	4	4	4.5	**28.5**	29.9	30.8	31.4	32.2	32.8	27.1	26.2	25.7	24.8	24.2
4.5	4.5	4.5	4.5	4.5	3	3	**28.5**										
4	4	4	4	4	4	5	**29.0**	30.5	31.3	31.9	32.8	33.4	27.6	26.7	26.1	25.2	24.7
4	4	4	4	4	4.5	4.5	**29.0**										
3.5	3.5	3.5	3.5	5	5	5	**29.0**										
4	4	4	4	4.5	4.5	4.5	**29.5**	31.0	31.9	32.5	33.3	33.9	28.0	27.1	26.6	25.7	25.1
4.5	4.5	4.5	4.5	4	4	4	**30.0**	31.5	32.4	33.0	33.9	34.5	28.5	27.6	27.0	26.1	25.5
4	4	4	4	4	5	5	**30.0**										
4.5	4.5	4.5	4.5	4.5	4	4	**30.5**	32.0	32.9	33.6	34.5	35.1	29.0	28.1	27.5	26.5	25.9
4.5	4.5	4.5	4.5	4.5	4.5	4	**31.0**	32.6	33.5	34.1	35.0	35.7	29.5	28.5	27.9	27.0	26.4
4	4	4	4	5	5	5	**31.0**										
5	5	5	5	5	3	3	**31.0**										
4.5	4.5	4.5	4.5	4.5	4.5	4.5	**31.5**	33.1	34.0	34.7	35.6	36.2	29.9	29.0	28.4	27.4	26.8
4.5	4.5	4.5	4.5	4.5	4.5	5	**32.0**	33.6	34.6	35.2	36.2	36.8	30.4	29.4	28.8	27.8	27.2
5	5	5	5	4	4	4	**32.0**										
5	5	5	5	5	3.5	3.5	**32.0**										
4.5	4.5	4.5	4.5	4.5	5	5	**32.5**	34.1	35.1	35.8	36.7	37.4	30.9	29.9	29.3	28.3	27.6
4.5	4.5	4.5	4.5	5	5	5	**33.0**	34.7	35.6	36.3	37.3	38.0	31.4	30.4	29.7	28.7	28.1
5	5	5	5	5	5	3	**33.0**										
5	5	5	5	5	4	4	**33.0**										
5	5	5	5	4.5	4.5	4.5	**33.5**	35.2	36.2	36.9	37.9	38.5	31.8	30.8	30.2	29.1	28.5
5	5	5	5	5	5	4	**34.0**	35.7	36.7	37.4	38.4	39.1	32.3	31.3	30.6	29.6	28.9
5	5	5	5	5	4.5	4.5	**34.0**										
5	5	5	5	5	5	4.5	**34.5**	36.2	37.3	38.0	39.0	39.7	32.8	31.7	31.1	30.0	29.3
4.5	4.5	4.5	4.5	4.5	6	6	**34.5**										
5	5	5	5	5	5	5	**35.0**	36.8	37.8	38.5	39.6	40.3	33.3	32.2	31.5	30.5	29.8
5	5	5	5	5	5	5.5	**35.5**	37.3	38.3	39.1	40.1	40.8	33.7	32.7	32.0	30.9	30.2
5	5	5	5	5	5	6	**36.0**	37.8	38.9	39.6	40.7	41.4	34.2	33.1	32.4	31.3	30.6
5	5	5	5	5	5.5	5.5	**36.0**										
5	5	5	5	5.5	5.5	5.5	**36.5**	38.3	39.4	40.2	41.2	42.0	34.7	33.6	32.9	31.8	31.0
5.5	5.5	5.5	5.5	5	5	5	**37.0**	38.9	40.0	40.7	41.8	42.6	35.2	34.0	33.3	32.2	31.5
5.5	5.5	5.5	5.5	5.5	5	5	**37.5**	39.4	40.5	41.3	42.4	43.1	35.6	34.5	33.8	32.6	31.9
5.5	5.5	5.5	5.5	5.5	5.5	5	**38.0**	39.9	41.0	41.8	42.9	43.7	36.1	35.0	34.2	33.1	32.3
6	6	6	6	6	4	4	**38.0**										
5.5	5.5	5.5	5.5	5.5	5.5	5.5	**38.5**	40.4	41.6	42.4	43.5	44.3	36.6	35.4	34.7	33.5	32.7
6	6	6	6	5	5	5	**39.0**	41.0	42.1	42.9	44.1	44.9	37.1	35.9	35.1	33.9	33.2
5.5	5.5	5.5	5.5	5.5	5.5	6	**39.0**										
5.5	5.5	5.5	5.5	5.5	6	6	**39.5**	41.5	42.7	43.5	44.6	45.4	37.5	36.3	35.6	34.4	33.6
5.5	5.5	5.5	5.5	6	6	6	**40.0**	42.0	43.2	44.0	45.2	46.0	38.0	36.8	36.0	34.8	34.0
6	6	6	6	6	5	5	**40.0**										
6	6	6	6	6	6	4	**40.0**										

WARFARIN DOSING CHART

Mg							Total	Increase (+)					Reduce (-)				
Mon	Tue	Wed	Thu	Fri	Sat	Sun	Weekly	5%	8%	10%	13%	15%	5%	8%	10%	13%	15%
6	6	6	6	5.5	5.5	5.5	**40.5**	42.5	43.7	44.6	45.8	46.6	38.5	37.3	36.5	35.2	34.4
6	6	6	6	6	6	4.5	**40.5**										
6	6	6	6	6	6	5	**41.0**	43.1	44.3	45.1	46.3	47.2	39.0	37.7	36.9	35.7	34.9
6	6	6	6	6	5.5	5.5	**41.0**										
6	6	6	6	6	6	5.5	**41.5**	43.6	44.8	45.7	46.9	47.7	39.4	38.2	37.4	36.1	35.3
5.5	5.5	5.5	5.5	5.5	7	7	**41.5**										
6	6	6	6	6	6	6	**42.0**	44.1	45.4	46.2	47.5	48.3	39.9	38.6	37.8	36.5	35.7
6	6	6	6	6	6	6.5	**42.5**	44.6	45.9	46.8	48.0	48.9	40.4	39.1	38.3	37.0	36.1
6	6	6	6	6	6	7	**43.0**	45.2	46.4	47.3	48.6	49.5	40.9	39.6	38.7	37.4	36.6
6	6	6	6	6	6.5	6.5	**43.0**	45.2	46.4	47.3	48.6	49.5	40.9	39.6	38.7	37.4	36.6
6	6	6	6	6.5	6.5	6.5	**43.5**	45.7	47.0	47.9	49.2	50.0	41.3	40.0	39.2	37.8	37.0
6	6	6	6	6	7	7	**44.0**	46.2	47.5	48.4	49.7	50.6	41.8	40.5	39.6	38.3	37.4
6.5	6.5	6.5	6.5	6.5	6	6	**44.5**	46.7	48.1	49.0	50.3	51.2	42.3	40.9	40.1	38.7	37.8
6.5	6.5	6.5	6.5	6.5	6.5	6	**45.0**	47.3	48.6	49.5	50.9	51.8	42.8	41.4	40.5	39.2	38.3
7	7	7	7	7	5	5	**45.0**										
6.5	6.5	6.5	6.5	6.5	6.5	6.5	**45.5**	47.8	49.1	50.1	51.4	52.3	43.2	41.9	41.0	39.6	38.7
6.5	6.5	6.5	6.5	6.5	6.5	7	**46.0**	48.3	49.7	50.6	52.0	52.9	43.7	42.3	41.4	40.0	39.1
6.5	6.5	6.5	6.5	6.5	7	7	**46.5**	48.8	50.2	51.2	52.5	53.5	44.2	42.8	41.9	40.5	39.5
6.5	6.5	6.5	6.5	7	7	7	**47.0**	49.4	50.8	51.7	53.1	54.1	44.7	43.2	42.3	40.9	40.0
7	7	7	7	7	7	5	**47.0**										
7	7	7	7	6.5	6.5	6.5	**47.5**	49.9	51.3	52.3	53.7	54.6	45.1	43.7	42.8	41.3	40.4
7	7	7	7	7	7	6	**48.0**	50.4	51.8	52.8	54.2	55.2	45.6	44.2	43.2	41.8	40.8
7	7	7	7	7	6.5	6.5	**48.0**										
7	7	7	7	7	7	6.5	**48.5**	50.9	52.4	53.4	54.8	55.8	46.1	44.6	43.7	42.2	41.2
6.5	6.5	6.5	6.5	6.5	8	8	**48.5**										
7	7	7	7	7	7	7	**49.0**	51.5	52.9	53.9	55.4	56.4	46.6	45.1	44.1	42.6	41.7
7	7	7	7	7	7	7.5	**49.5**	52.0	53.5	54.5	55.9	56.9	47.0	45.5	44.6	43.1	42.1
7	7	7	7	7	7	8	**50.0**	52.5	54.0	55.0	56.5	57.5	47.5	46.0	45.0	43.5	42.5
7	7	7	7	7	7.5	7.5	**50.0**										
7.5	7.5	7.5	7.5	7.5	7.5	5	**50.0**										
7	7	7	7	7.5	7.5	7.5	**50.5**	53.0	54.5	55.6	57.1	58.1	48.0	46.5	45.5	43.9	42.9
7	7	7	7	7	8	8	**51.0**	53.6	55.1	56.1	57.6	58.7	48.5	46.9	45.9	44.4	43.4
7.5	7.5	7.5	7.5	7	7	7	**51.0**										
7.5	7.5	7.5	7.5	7.5	7	7	**51.5**	54.1	55.6	56.7	58.2	59.2	48.9	47.4	46.4	44.8	43.8
7.5	7.5	7.5	7.5	7.5	7.5	7	**52.0**	54.6	56.2	57.2	58.8	59.8	49.4	47.8	46.8	45.2	44.2
7.5	7.5	7.5	7.5	7.5	7.5	7.5	**52.5**	55.1	56.7	57.8	59.3	60.4	49.9	48.3	47.3	45.7	44.6
7.5	7.5	7.5	7.5	7.5	7.5	8	**53.0**	55.7	57.2	58.3	59.9	61.0	50.4	48.8	47.7	46.1	45.1
7.5	7.5	7.5	7.5	7.5	8	8	**53.5**	56.2	57.8	58.9	60.5	61.5	50.8	49.2	48.2	46.5	45.5
7.5	7.5	7.5	7.5	8	8	8	**54.0**	56.7	58.3	59.4	61.0	62.1	51.3	49.7	48.6	47.0	45.9
7.5	7.5	7.5	7.5	7.5	7.5	9	**54.0**										
8	8	8	8	8	8	7	**55.0**	57.8	59.4	60.5	62.2	63.3	52.3	50.6	49.5	47.9	46.8
8	8	8	8	8	7.5	7.5	**55.0**										
8	8	8	8	8	8	7.5	**55.5**	58.3	59.9	61.1	62.7	63.8	52.7	51.1	50.0	48.3	47.2
8	8	8	8	8	8	8	**56.0**	58.8	60.5	61.6	63.3	64.4	53.2	51.5	50.4	48.7	47.6
8	8	8	8	8	8	8.5	**56.5**	59.3	61.0	62.2	63.8	65.0	53.7	52.0	50.9	49.2	48.0
8	8	8	8	8	8	9	**57.0**	59.9	61.6	62.7	64.4	65.6	54.2	52.4	51.3	49.6	48.5
8	8	8	8	8	8.5	8.5	**57.0**										
8	8	8	8	8.5	8.5	8.5	**57.5**	60.4	62.1	63.3	65.0	66.1	54.6	52.9	51.8	50.0	48.9
8	8	8	8	8	9	9	**58.0**	60.9	62.6	63.8	65.5	66.7	55.1	53.4	52.2	50.5	49.3
8.5	8.5	8.5	8.5	8.5	8	8	**58.5**	61.4	63.2	64.4	66.1	67.3	55.6	53.8	52.7	50.9	49.7
8.5	8.5	8.5	8.5	8.5	8.5	8	**59.0**	62.0	63.7	64.9	66.7	67.9	56.1	54.3	53.1	51.3	50.2
8	8	8	8	9	9	9	**59.0**										

WARFARIN DOSING CHART																	
Mg							Total	Increase (+)					Reduce (-)				
Mon	Tue	Wed	Thu	Fri	Sat	Sun	Weekly	5%	8%	10%	13%	15%	5%	8%	10%	13%	15%
8.5	8.5	8.5	8.5	8.5	8.5	8.5	**59.5**	62.5	64.3	65.5	67.2	68.4	56.5	54.7	53.6	51.8	50.6
8.5	8.5	8.5	8.5	8.5	8.5	9	**60.0**	63.0	64.8	66.0	67.8	69.0	57.0	55.2	54.0	52.2	51.0
8.5	8.5	8.5	8.5	8.5	9	9	**60.5**	63.5	65.3	66.6	68.4	69.6	57.5	55.7	54.5	52.6	51.4
8.5	8.5	8.5	8.5	9	9	9	**61.0**	64.1	65.9	67.1	68.9	70.2	58.0	56.1	54.9	53.1	51.9
9	9	9	9	8.5	8.5	8.5	**61.5**	64.6	66.4	67.7	69.5	70.7	58.4	56.6	55.4	53.5	52.3
9	9	9	9	9	9	8	**62.0**	65.1	67.0	68.2	70.1	71.3	58.9	57.0	55.8	53.9	52.7
9	9	9	9	9	8.5	8.5	**62.0**	65.1	67.0	68.2	70.1	71.3	58.9	57.0	55.8	53.9	52.7
9	9	9	9	9	9	8.5	**62.5**	65.6	67.5	68.8	70.6	71.9	59.4	57.5	56.3	54.4	53.1
9	9	9	9	9	9	9	**63.0**	66.2	68.0	69.3	71.2	72.5	59.9	58.0	56.7	54.8	53.6
9	9	9	9	9	9	9.5	**63.5**	66.7	68.6	69.9	71.8	73.0	60.3	58.4	57.2	55.2	54.0
9	9	9	9	9	9.5	9.5	**64.0**	67.2	69.1	70.4	72.3	73.6	60.8	58.9	57.6	55.7	54.4
9	9	9	9	9	9	10	**64.0**										
9	9	9	9	9.5	9.5	9.5	**64.5**	67.7	69.7	71.0	72.9	74.2	61.3	59.3	58.1	56.1	54.8
9	9	9	9	9	10	10	**65.0**	68.3	70.2	71.5	73.5	74.8	61.8	59.8	58.5	56.6	55.3
9.5	9.5	9.5	9.5	9.5	9	9	**65.5**	68.8	70.7	72.1	74.0	75.3	62.2	60.3	59.0	57.0	55.7
9.5	9.5	9.5	9.5	9.5	9.5	9	**66.0**	69.3	71.3	72.6	74.6	75.9	62.7	60.7	59.4	57.4	56.1
9.5	9.5	9.5	9.5	9.5	9.5	9.5	**66.5**	69.8	71.8	73.2	75.1	76.5	63.2	61.2	59.9	57.9	56.5
9.5	9.5	9.5	9.5	9.5	9.5	10	**67.0**	70.4	72.4	73.7	75.7	77.1	63.7	61.6	60.3	58.3	57.0
9.5	9.5	9.5	9.5	9.5	10	10	**67.5**	70.9	72.9	74.3	76.3	77.6	64.1	62.1	60.8	58.7	57.4
9.5	9.5	9.5	9.5	10	10	10	**68.0**	71.4	73.4	74.8	76.8	78.2	64.6	62.6	61.2	59.2	57.8
9.5	9.5	9.5	9.5	9.5	9.5	11	**68.0**										
10	10	10	10	9.5	9.5	9.5	**68.5**	71.9	74.0	75.4	77.4	78.8	65.1	63.0	61.7	59.6	58.2
10	10	10	10	10	9.5	9.5	**69.0**	72.5	74.5	75.9	78.0	79.4	65.6	63.5	62.1	60.0	58.7
10	10	10	10	10	10	9	**69.0**										
10	10	10	10	10	10	9.5	**69.5**	73.0	75.1	76.5	78.5	79.9	66.0	63.9	62.6	60.5	59.1
10	10	10	10	10	10	10	**70.0**	73.5	75.6	77.0	79.1	80.5	66.5	64.4	63.0	60.9	59.5
10	10	10	10	10	10	11	**70.5**	74.0	76.1	77.6	79.7	81.1	67.0	64.9	63.5	61.3	59.9
10	10	10	10	10	11	11	**71.0**	74.6	76.7	78.1	80.2	81.7	67.5	65.3	63.9	61.8	60.4
10	10	10	10	10	10	11	**71.0**										
10	10	10	10	11	11	11	**71.5**	75.1	77.2	78.7	80.8	82.2	67.9	65.8	64.4	62.2	60.8
10	10	10	10	10	11	11	**72.0**	75.6	77.8	79.2	81.4	82.8	68.4	66.2	64.8	62.6	61.2
10	10	10	10	10	10	12	**72.0**										
11	11	11	11	11	10	10	**72.5**	76.1	78.3	79.8	81.9	83.4	68.9	66.7	65.3	63.1	61.6
11	11	11	11	11	11	10	**73.0**	76.7	78.8	80.3	82.5	84.0	69.4	67.2	65.7	63.5	62.1
10	10	10	10	10	12	12	**73.0**										
10	10	10	10	10	10	13	**73.0**										
11	11	11	11	11	11	11	**73.5**	77.2	79.4	80.9	83.1	84.5	69.8	67.6	66.2	63.9	62.5
11	11	11	11	10	10	10	**74.0**	77.7	79.9	81.4	83.6	85.1	70.3	68.1	66.6	64.4	62.9
11	11	11	11	11	11	11	**74.5**	78.2	80.5	82.0	84.2	85.7	70.8	68.5	67.1	64.8	63.3
11	11	11	11	11	10	10	**75.0**	78.8	81.0	82.5	84.8	86.3	71.3	69.0	67.5	65.3	63.8
11	11	11	11	11	11	11	**75.0**										
11	11	11	11	11	11	11	**75.5**	79.3	81.5	83.1	85.3	86.8	71.7	69.5	68.0	65.7	64.2
11	11	11	11	11	11	10	**76.0**	79.8	82.1	83.6	85.9	87.4	72.2	69.9	68.4	66.1	64.6
10	10	10	10	12	12	12	**76.0**										
11	11	11	11	11	11	11	**76.5**	80.3	82.6	84.2	86.4	88.0	72.7	70.4	68.9	66.6	65.0
11	11	11	11	11	11	11	**77.0**	80.9	83.2	84.7	87.0	88.6	73.2	70.8	69.3	67.0	65.5
11	11	11	11	11	11	12	**77.5**	81.4	83.7	85.3	87.6	89.1	73.6	71.3	69.8	67.4	65.9
11	11	11	11	11	12	12	**78.0**	81.9	84.2	85.8	88.1	89.7	74.1	71.8	70.2	67.9	66.3
11	11	11	11	11	11	12	**78.0**										
11	11	11	11	12	12	12	**78.5**	82.4	84.8	86.4	88.7	90.3	74.6	72.2	70.7	68.3	66.7
11	11	11	11	11	12	12	**79.0**	83.0	85.3	86.9	89.3	90.9	75.1	72.7	71.1	68.7	67.2
12	12	12	12	12	11	11	**79.5**	83.5	85.9	87.5	89.8	91.4	75.5	73.1	71.6	69.2	67.6

WARFARIN DOSING CHART

Mg							Total	Increase (+)					Reduce (-)				
Mo	Tue	Wed	Thu	Fri	Sat	Sun	Weekly	5%	8%	10%	13%	15%	5%	8%	10%	13%	15%
11	11	11	11	12	12	12	**80.0**	84.0	86.4	88.0	90.4	92.0	76.0	73.6	72.0	69.6	68.0
12	12	12	12	12	12	11	**80.0**										
12	12	12	12	12	12	12	**80.5**	84.5	86.9	88.6	91.0	92.6	76.5	74.1	72.5	70.0	68.4
12	12	12	12	12	12	12	**81.0**	85.1	87.5	89.1	91.5	93.2	77.0	74.5	72.9	70.5	68.9
11	11	11	11	13	13	13	**81.0**										
12	12	12	12	12	12	12	**81.5**	85.6	88.0	89.7	92.1	93.7	77.4	75.0	73.4	70.9	69.3
12	12	12	12	12	12	12	**82.0**	86.1	88.6	90.2	92.7	94.3	77.9	75.4	73.8	71.3	69.7
12	12	12	12	12	11	11	**82.0**	86.1	88.6	90.2	92.7	94.3	77.9	75.4	73.8	71.3	69.7
12	12	12	12	12	12	12	**82.5**	86.6	89.1	90.8	93.2	94.9	78.4	75.9	74.3	71.8	70.1
12	12	12	12	12	12	11	**83.0**	87.2	89.6	91.3	93.8	95.5	78.9	76.4	74.7	72.2	70.6
12	12	12	12	12	12	12	**83.0**										
12	12	12	12	12	12	12	**83.5**	87.7	90.2	91.9	94.4	96.0	79.3	76.8	75.2	72.6	71.0
12	12	12	12	12	12	12	**84.0**	88.2	90.7	92.4	94.9	96.6	79.8	77.3	75.6	73.1	71.4
12	12	12	12	12	12	13	**84.5**	88.7	91.3	93.0	95.5	97.2	80.3	77.7	76.1	73.5	71.8
12	12	12	12	12	13	13	**85.0**	89.3	91.8	93.5	96.1	97.8	80.8	78.2	76.5	74.0	72.3
12	12	12	12	12	12	13	**85.0**										
13	13	13	13	11	11	11	**85.0**										
12	12	12	12	13	13	13	**85.5**	89.8	92.3	94.1	96.6	98.3	81.2	78.7	77.0	74.4	72.7
13	13	13	13	12	12	12	**86.0**	90.3	92.9	94.6	97.2	98.9	81.7	79.1	77.4	74.8	73.1
12	12	12	12	12	13	13	**86.0**										
13	13	13	13	13	12	12	**86.5**	90.8	93.4	95.2	97.7	99.5	82.2	79.6	77.9	75.3	73.5
13	13	13	13	13	13	12	**87.0**	91.4	94.0	95.7	98.3	100.1	82.7	80.0	78.3	75.7	74.0
12	12	12	12	13	13	13	**87.0**										
13	13	13	13	13	13	13	**87.5**	91.9	94.5	96.3	98.9	100.6	83.1	80.5	78.8	76.1	74.4
13	13	13	13	12	12	12	**88.0**	92.4	95.0	96.8	99.4	101.2	83.6	81.0	79.2	76.6	74.8
13	13	13	13	13	12	12	**88.0**										
13	13	13	13	13	13	10	**88.0**										
13	13	13	13	13	13	13	**88.5**	92.9	95.6	97.4	100.0	101.8	84.1	81.4	79.7	77.0	75.2
13	13	13	13	13	12	12	**89.0**	93.5	96.1	97.9	100.6	102.4	84.6	81.9	80.1	77.4	75.7
13	13	13	13	13	13	11	**89.0**										
13	13	13	13	13	13	13	**89.5**	94.0	96.7	98.5	101.1	102.9	85.0	82.3	80.6	77.9	76.1
13	13	13	13	13	13	13	**90.0**										
13	13	13	13	13	13	12	**90.0**	94.5	97.2	99.0	101.7	103.5	85.5	82.8	81.0	78.3	76.5
13	13	13	13	13	13	13	**90.5**	95.0	97.7	99.6	102.3	104.1	86.0	83.3	81.5	78.7	76.9
13	13	13	13	13	13	13	**91.0**	95.6	98.3	100.1	102.8	104.7	86.5	83.7	81.9	79.2	77.4
13	13	13	13	13	13	14	**91.5**	96.1	98.8	100.7	103.4	105.2	86.9	84.2	82.4	79.6	77.8
13	13	13	13	13	13	14	**92.0**	96.6	99.4	101.2	104.0	105.8	87.4	84.6	82.8	80.0	78.2
13	14	13	14	13	14	13	**92.5**	97.1	99.9	101.8	104.5	106.4	87.9	85.1	83.3	80.5	78.6
13	13	13	13	13	14	14	**93.0**	97.7	100.4	102.3	105.1	107.0	88.4	85.6	83.7	80.9	79.1
14	14	14	14	14	13	13	**93.5**	98.2	101.0	102.9	105.7	107.5	88.8	86.0	84.2	81.3	79.5
13	13	13	13	14	14	14	**94.0**	98.7	101.5	103.4	106.2	108.1	89.3	86.5	84.6	81.8	79.9
14	14	14	14	14	14	14	**94.5**	99.2	102.1	104.0	106.8	108.7	89.8	86.9	85.1	82.2	80.3
14	14	14	14	13	13	13	**95.0**	99.8	102.6	104.5	107.4	109.3	90.3	87.4	85.5	82.7	80.8
14	14	14	14	14	14	14	**95.5**	100.3	103.1	105.1	107.9	109.8	90.7	87.9	86.0	83.1	81.2
14	14	14	14	14	13	13	**96.0**	100.8	103.7	105.6	108.5	110.4	91.2	88.3	86.4	83.5	81.6
14	14	14	14	14	14	14	**96.5**	101.3	104.2	106.2	109.0	111.0	91.7	88.8	86.9	84.0	82.0
14	14	14	14	14	14	13	**97.0**	101.9	104.8	106.7	109.6	111.6	92.2	89.2	87.3	84.4	82.5
14	14	14	14	14	14	14	**97.5**	102.4	105.3	107.3	110.2	112.1	92.6	89.7	87.8	84.8	82.9
14	14	14	14	14	14	14	**98.0**	102.9	105.8	107.8	110.7	112.7	93.1	90.2	88.2	85.3	83.3
14	14	14	14	14	14	15	**98.5**	103.4	106.4	108.4	111.3	113.3	93.6	90.6	88.7	85.7	83.7
14	14	14	14	14	14	15	**99.0**	104.0	106.9	108.9	111.9	113.9	94.1	91.1	89.1	86.1	84.2
14	14	14	14	15	15	15	**99.5**	104.5	107.5	109.5	112.4	114.4	94.5	91.5	89.6	86.6	84.6
14	14	14	14	14	15	15	**100.0**	105.0	108.0	110.0	113.0	115.0	95.0	92.0	90.0	87.0	85.0
15	15	15	15	15	14	14	**100.5**	105.5	108.5	110.6	113.6	115.6	95.5	92.5	90.5	87.4	85.4
14	14	14	14	15	15	15	**101.0**	106.1	109.1	111.1	114.1	116.2	96.0	92.9	90.9	87.9	85.9
15	15	15	15	15	15	15	**101.5**	106.6	109.6	111.7	114.7	116.7	96.4	93.4	91.4	88.3	86.3
15	15	15	15	14	14	14	**102.0**	107.1	110.2	112.2	115.3	117.3	96.9	93.8	91.8	88.7	86.7

Dose Initiation

For patients newly initiated with warfarin – calculate weekly dose then increase dose according to Table 1 and to give appointment according to Table 2.

Table 1 : Warfarin Initiation for Day 1-7 (INR target 2-3)[3]

INR	**Non-sensitive Patients**	**Sensitive Patients**†
Initial dose	5 mg/ 5 mg then 3 mg daily	3 mg then 2 mg daily
Day 8 (INR)	**Dose adjustment**	
< 1.5	Increase weekly dose by 10-15%, review in 7 days	
1.5 – 1.9	Maintain dose, review in 7-14 days	
2.0 – 3.0	Reduce 0.5 or 1 mg from weekly dose, review in 7-10 days	
> 3.0	Hold and initiate at lower dose when INR in range, review in 7 days	

INR:international normalized ratio

†Sensitive Patients - elderly patients or in patients who are debilitated, malnourished, have congestive heart disease, have liver disease, have had recent major surgery, or are taking medications known to increase the sensitivity to warfarin (eg. amiodarone)[3]

Table 2 : Frequency of INR Monitoring[3]

INR monitoring	
INR	**Suggested approach**
Initiation of warfarin	Monitor every 3 - 7 days, then gradually increase to weekly, biweekly, monthly if stable
INR reaches target	Recheck 2 weeks x 2, then every 4 weeks if stable
INR therapeutic for 8-10 weeks consecutively	May increase interval to 6 - 12 weeks with high compliance and good patient education; increase frequency with illness, changes in medication, history of highly variable INR levels
INR outside target range +/- 0.2 points	Recheck in 1 - 2 weeks; if persists, adjust dose and recheck in 1 - 2 weeks
INR > from target range but < 5	Hold warfarin, adjust dose, recheck in 1 week

INR:international normalized ratio

Direct Oral Anticoagulants (DOAC)

Indications[1,4–7]

Drug	Indications*
Dabigatran	- **To reduce the risk of stroke and systemic embolism in adult patients with non-valvular atrial fibrillation** - **For the treatment of deep venous thrombosis (DVT) and pulmonary embolism (PE) in adult patients who have been treated with a parenteral anticoagulant for 5-10 days** - **To reduce the risk of recurrence of DVT and PE in adult patients who have been previously treated** - **For the prophylaxis of DVT and PE in adult patients who have undergone hip replacement surgery** - For the treatment of venous thromboembolic events (VTE) in paediatric patients 8 to less than 18 years of age who have been treated with a parenteral anticoagulant for at least 5 days - To reduce the risk of recurrence of VTE in paediatric patients 8 to less than 18 years of age who have been previously treated
Rivaroxaban	- **To reduce risk of stroke and systemic embolism in nonvalvular atrial fibrillation** - **For treatment of deep vein thrombosis (DVT)** - **For treatment of pulmonary embolism (PE)** - **For reduction in the risk of recurrence of DVT or PE** - **For the prophylaxis of DVT, which may lead to PE in patients undergoing knee or hip replacement surgery** - For prophylaxis of venous thromboembolism (VTE) in acutely ill medical patients - To reduce the risk of major cardiovascular events in patients with coronary artery disease (CAD) - To reduce the risk of major thrombotic vascular events in patients with peripheral artery disease (PAD), including patients after recent lower extremity revascularization due to symptomatic PAD - For treatment of VTE and reduction in the risk of recurrent VTE in paediatric patients from birth to less than 18 years - For thromboprophylaxis in paediatric patients 2 years and older with congenital heart disease after the Fontan procedure
Apixaban	- **To reduce the risk of stroke and systemic embolism in patients with nonvalvular atrial fibrillation**
Edoxaban	- To reduce the risk of stroke and systemic embolism (SE) in patients with nonvalvular atrial fibrillation (NVAF)

	Limitation of use : should not be used in patients with creatinine clearance (CrCL) > 95 mL/min because of increased risk of ischemic stroke compared to warfarin at the highest dose studied (60 mg) - For the treatment of deep vein thrombosis (DVT) and pulmonary embolism (PE) following 5-10 days of initial therapy with a parenteral anticoagulant

*Indications obtained from the Prescribing Information of individual drugs. Indications in bold as per Ministry of Health Malaysia Drug Formulary (accessed from Intranet Pharmacy 20 July 2023).

Contraindications / Avoidance of DOAC : General[4–6]

Contraindications / Avoidance of DOAC : General
1. Moderate – severe mitral stenosis (MVA < 1.5 cm^2)
2. Mechanical prosthetic valve
3. Significant renal disease or dysfunction or ESRD or requiring dialysis - CrCl < 15 mL/min : avoid rivaroxaban, apixaban & edoxaban - CrCl < 30 mL/min : avoid dabigatran
4. Active bleeding or coagulopathy or prolonged APTT or INR
5. Concomitant with other anticoagulant or presence of significant therapeutic level or other anticoagulant or thrombolytic agent
6. Thrombocytopenia (<20,000 - 50,000*)
7. Allergic to DOAC
8. Liver failure or advanced liver disease
9. Antiphospholipid syndrome (triple positive)
10. Pregnancy and breastfeeding
11. Gastrointestinal & genitourinary malignancy

MVA:mitral valve area; ESRD:end stage renal disease; CrCl:creatinine clearance, APTT:activated partial thromboplastin time; INR:international normalized ratio; DOAC:direct oral anticoagulant

* Some will only avoid DOAC when < 20,000

DOAC Dosing in Venous Thromboembolism (VTE) [Deep Vein Thrombosis (DVT)/ Pulmonary Embolism (PE)][4–6]

Drug	Initiation	Maintenance dose
Dabigatran	LMWH 5 Days	150 mg BD or 110 mg BD
Rivaroxaban	15 mg BD 21 Days	20mg OD
Apixaban	10 mg BD 7 Days	5 mg BD
Edoxaban	LMWH 5 Days	60 mg OD (Weight > 60kg) 30 mg OD (Weight ≤ 60kg)

LMWH:low molecular weight heparin; BD:twice daily; OD:once daily

DOAC Dosing in Obese Patients with AF[8]

Use of DOAC in Obese Patients

BMI ≤ 30 kg/m^2	BMI 30 – 40 kg/m^2	BMI 40 - 49 kg/m^2	BMI ≥ 50 kg/m^2
↓	↓	↓	↓
Use any DOAC as per guidelines	Use any DOAC as per guidelines	Use warfarin Can consider **apixaban** or **edoxaban***	Use **warfarin** Very limited data on DOAC use

*Recommendation based on lack of specific evidence for dabigatran and rivaroxaban in this BMI category and sub-group analysis with apixaban and edoxaban not suggesting any inferior benefit compared to warfarin

BMI:body mass index; DOAC:direct oral anticoagulant

DOAC Dosing in Obese Patients with VTE[9]

Summary Guidance Statements for use of DOACs in Patients with Obesity

1) Consistent with the 2016 International Society on Thrombosis and Haemostasis Scientific and Standardization Committee (ISTH SSC) recommendations, we conclude that the use of any DOAC is appropriate for patients with BMI up to 40 kg/m^2 or weight 120 kg. For patients with BMI >40 kg/m^2 or weight >120 kg, we recommend that the individual DOACs should be used as follows :

2) For treatment of VTE, we suggest that standard doses of rivaroxaban or apixaban are among appropriate anticoagulant options regardless of high BMI and weight. Fewer supportive data exist for apixaban than rivaroxaban. VKA, weight-based LMWH (per manufacturers' recommendations), and fondaparinux are also options.

3) For primary prevention of VTE, we suggest that standard doses of rivaroxaban or apixaban are among appropriate anticoagulant options regardless of high BMI and weight. Drug approval is restricted to elective hip and knee arthroplasty and (in some countries) extended VTE prevention following acute medical illness.

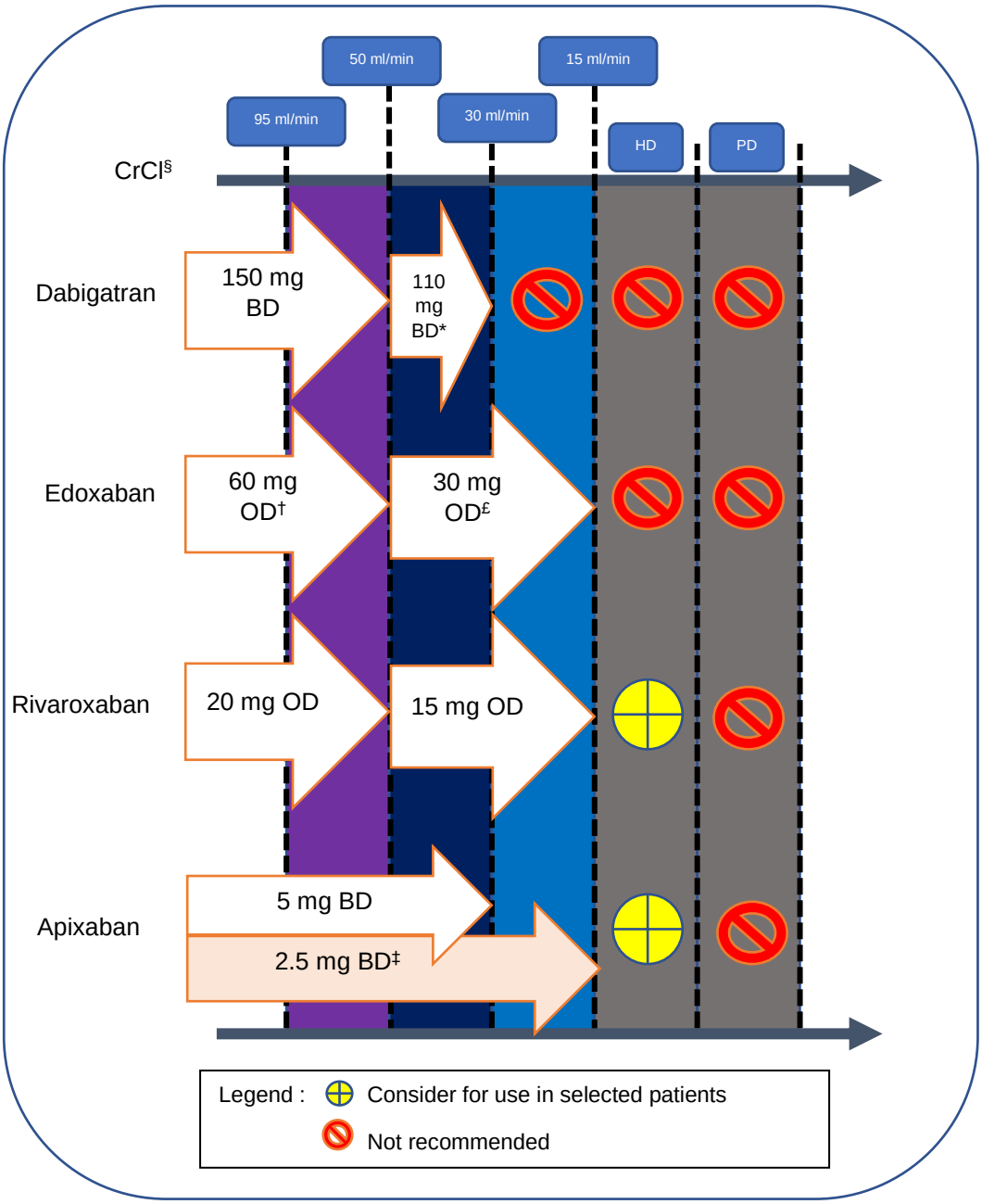
4)We suggest not to use dabigatran, edoxaban or betrixaban for VTE treatment and prevention in patients with BMI >40 kg/m^2 or weight >120 kg, given unconvincing data for dabigatran, and lack of clinical or PK/PD data for edoxaban and betrixaban.

5)We suggest not to regularly follow peak or trough drug-specific DOAC levels because there are insufficient data to influence management decisions.

6)We suggest not to use DOAC for treatment or prevention of VTE in the acute setting after bariatric surgery (because of concerns of decreased absorption), and instead, to initiate such patients on parenteral anticoagulation in the early postsurgical phase. We suggest that switching to VKA or DOAC may be considered after at least 4 weeks of parenteral treatment, and if so, suggest obtaining a DOAC trough level to check for drug absorption and bioavailability.

BMI:body mass index; DOAC:direct oral anticoagulant; LMWH:low molecular weight heparin; VKA:vitamin K antagonist; VTE:venous thromboembolism

DOAC Dosing for AF with Respect to Renal Function [10]



CrCL:creatinine clearance; HD:haemodialysis; PD:peritoneal dialysis; BD:twice daily; OD:daily

§Creatinine clearance estimated by the Cockgroft-Gault equation

*150 mg twice daily may be considered in suitable patients with low bleeding risk

†Should not be used in patients with CrCL > 95 mL/min because of increased risk of ischemic stroke compared to warfarin at the highest dose studied (60 mg)

£If used for the treatment of DVT or PE, recommended dose is 30 mg OD for patients with CrCL 15 - 50 mL/min or body weight ≤60 kg or who use certain P-gp inhibitors(verapamil, quinidine, erythromycin, ketoconazole etc.)

‡2.5 mg twice daily if at least two out of three criteria : age ≥ 80 years, body weight ≤ 60 kg, creatinine ≥ 1.5 mg/dL (133 µmol/L). In patients with CrCl 15-29 ml/min, recommended dose is 2.5 mg twice daily independent of age or body weight

Unfractionated Heparin, Low Molecular Weight Heparin and Factor Xa Inhibitor Dosing for VTE[11]

Drug	Dosage
Heparin	80 IU/kg IV (max 10,000 U) for the bolus dose followed by a continuous infusion of 18 IU/kg/h. The aim of therapy is to achieve an aPTT of 1.5 – 2x baseline
Dalteparin sodium	200 U/kg/day SC divided qd-bid. Max : 18,000 U/dose. Continue for > 5 days and overlap with warfarin until INR 2 - 3
Enoxaparin sodium	1 mg/kg SC q12h. Alt: 1.5 mg/kg SC qd. Continue for >5 days and overlap with warfarin until INR 2 - 3
Fondaparinux	5 mg (body weight < 50kg), 7.5 mg (50-100 kg), or 10 mg (>100 kg) SC once daily

Alt:alternative; aPTT:activated partial thromboplastin time; INR:international normalized ratio; IV:intravenous; max:maximum; qd:daily; bid:twice a day; SC:subcutaneous

Enoxaparin Dosing[12]

Indication	Renal function	Weight			
		< 50 kg	50-100 kg	100-150 kg	> 150 kg
Thromboprophylaxis	CrCl ≥ 30mL/min	20 mg once daily	40 mg once daily	40 mg twice a day	60 mg twice a day
	CrCl 15-30 mL/min	20 mg once daily	20 mg once daily	40 mg once daily	60 mg once daily
	CrCl < 15 mL/min	20 mg once daily (this is considered generally safe in this patient group but consider seeking specialist advice if used for prolonged periods and/or the patient is at extremes of weight (anti-Xa monitoring can be considered)			
DVT or PE treatment	CrCl ≥ 30mL/min	1.5 mg/kg once daily			1 mg/kg twice a day
	CrCl 15-30 mL/min	1 mg/kg once daily			
	CrCl < 15 mL/min	Seek specialist advice			
Mechanical heart valves if INR low[1]	CrCl ≥ 30mL/min	1 mg/kg twice a day			
	CrCl 15-30 mL/min	1 mg/kg once daily			
	CrCl < 15 mL/min	Seek specialist advice			
[1]Mechanical heart valves if INR < 1.8 (unless INR range 1.5 – 2.5, then if < 1.5) and continue until INR at least > 2					

CrCl:creatinine clearance; DVT:deep vein thrombosis; INR:international normalized ratio; PE:pulmonary embolism

Conversion of Anticoagulants

From Warfarin or Injectable Anticoagulant to DOAC[4–7]

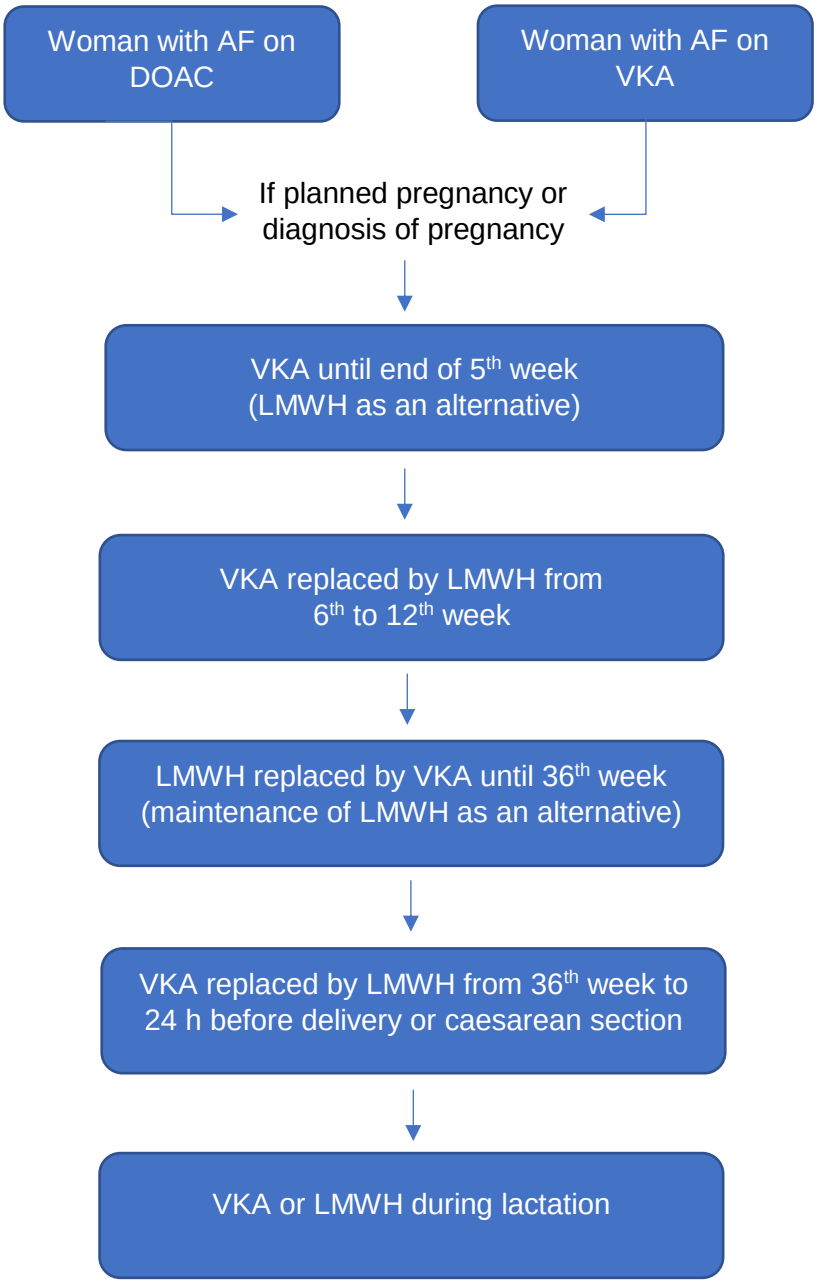
<table>
<tr><th>Switching</th><th>Warfarin to DOAC</th><th>LMWH to DOAC</th><th>UFH to DOAC</th><th>DOAC to another DOAC</th></tr>
<tr><td>Dabigatran</td><td>Start when INR < 2.0</td><td rowspan="2">Stop LMWH; Start DOAC ≤ 2 hours prior to the time of the next scheduled dose of LMWH</td><td rowspan="2">Start DOAC immediately after stopping IV UFH</td><td rowspan="4">Stop current DOAC; at time of next dose of current DOAC, start new DOAC</td></tr>
<tr><td>Rivaroxaban</td><td>Start when INR < 3.0</td></tr>
<tr><td>Apixaban</td><td>Start when INR < 2.0</td><td rowspan="2">Stop LMWH; start DOAC at the time when the next dose of LMWH is due</td></tr>
<tr><td>Edoxaban</td><td>Start when INR ≤ 2.5</td><td>Start edoxaban 4 hours after stopping IV UFH</td></tr>
<tr><td colspan="5">DOAC:direct oral anticoagulant; INR:international normalized ratio; IV:intravenous; LMWH:low molecular weight heparin; UFH:unfractionated heparin</td></tr>
</table>

From DOAC to Warfarin[4–7]

<table>
<tr><th>DOAC</th><th>Procedure</th></tr>
<tr><td>Dabigatran</td><td>Start warfarin as follows :
• For CrCl ≥ 50 mL/min, start warfarin 3 days before discontinuing dabigatran
• For CrCl 30-50 mL/min, start warfarin 2 days before discontinuing dabigatran
• For CrCl 15-30 mL/min, start warfarin 1 day before discontinuing dabigatran
• CrCl < 15 mL/min, no recommendations can be made</td></tr>
<tr><td>Rivaroxaban</td><td>Stop DOAC; start warfarin and LMWH at time of next scheduled DOAC dose*</td></tr>
<tr><td>Apixaban</td><td>Stop DOAC; start warfarin and LMWH at time of next scheduled DOAC dose and bridge until INR reaches an acceptable range</td></tr>
<tr><td>Edoxaban</td><td>• For 60 mg dose, reduce dose to 30 mg and start warfarin concomitantly
• For 30 mg dose, reduce dose to 15 mg and start warfarin concomitantly
• Stop edoxaban when INR ≥ 2.0</td></tr>
<tr><td colspan="2">CrCl:creatinine clearance; DOAC:direct oral anticoagulant; INR:international normalized ratio; LMWH:low molecular weight heparin
*No clinical trial data is available to guide converting patients from XARELTO® (Rivaroxaban) to warfarin. XARELTO® affects INR, so INR measurements made during coadministration with warfarin may not be useful for determining the appropriate dose of warfarin[13].</td></tr>
</table>

Anticoagulants in Pregnancy

Management of Anticoagulant Therapy during Pregnancy (Atrial Fibrillation) [14,15]



AF:atrial fibrillation; DOAC:direct oral anticoagulant; LMWH:low molecular weight heparin; VKA:vitamin K antagonist

Management of Anticoagulant Therapy during Pregnancy (Valve Prosthesis)[14]

Stage of pregnancy	Recommendation
First trimester	After discussion of risks vs benefits, consider discontinuing VKA between weeks 6-12 of gestation and replacing with : Adjusted-dose UFH with aPTT ≥ twice the control OR Twice daily adjusted-dose LMWH with target anti-Xa levels 4-6 hours post-dose of 0.8 – 1.2 U/mL Should especially consider if warfarin dose > 5 mg/day
	After discussion of risks vs benefits, should consider continuing VKA during first trimester if warfarin dose is < 5 mg/day
2nd – 3rd trimester	VKA recommended until the 36th week Low dose aspirin (75 – 100 mg) once daily is recommended for mechanical or bioprosthetic valves
	In patients choosing LMWH, post-dose anti-Xa* levels should be assessed weekly
	Any changes in anticoagulation should be carried out in a hospital setting
Beyond 36 weeks	VKA should be replaced at 36 weeks gestation with either : Adjusted-dose LMWH with target anti-Xa* levels 4-6 hours post-dose of 0.8 – 1.2 U/mL OR Dose-adjusted UFH with aPTT ≥ twice the control
	LMWH replaced with UFH at least 36 hours prior to delivery, continued until 4-6 hours before delivery and restarted 4-6 hours following delivery barring complications
	Caesarean delivery indicated in patients not switched off VKA

aPTT:activated partial thromboplastin time; LMWH: low molecular weight heparin; VKA:vitamin K antagonist; UFH:unfractionated heparin

*Anti-Xa level testing may not be available in some facilities

Management of Anticoagulant Therapy during Pregnancy and Postpartum Period (VTE)[16]



Acute VTE before the 37th week of gestation

Start full-dose weight-adjusted LMWH

once or twice daily dosing regimen

Continue treatment with full-dose LMWH

switch to twice daily dosing perinatally

Stop LMWH

with the onset of labour or 24 h before planned delivery

Resume therapy

6-12 h after vaginal delivery or 12-24 h after caesarean section

consider switch to warfarin

Acute proximal DVT or PE after the 37th week of gestation

Start full-dose weight-adjusted LMWH

twice daily dosing regimen

Switch to aPTT-adjusted UFH infusion perinatally†,§

consider planned delivery

Stop UFH

4-6 h before planned delivery or reduce infusion rate and stop UFH only in the ejection phase

Restart UFH infusion

4-6 h after delivery

Switch to LMWH when bleeding has subsided

Acute VTE in the postpartum period

Start full-dose weight-adjusted LMWH

once or twice daily dosing regimen

Switch to warfarin in lactating women

after an overlapping phase until INR is in target range

continue anticoagulation for a minimum total duration of 3 months and at least 6 weeks postpartum

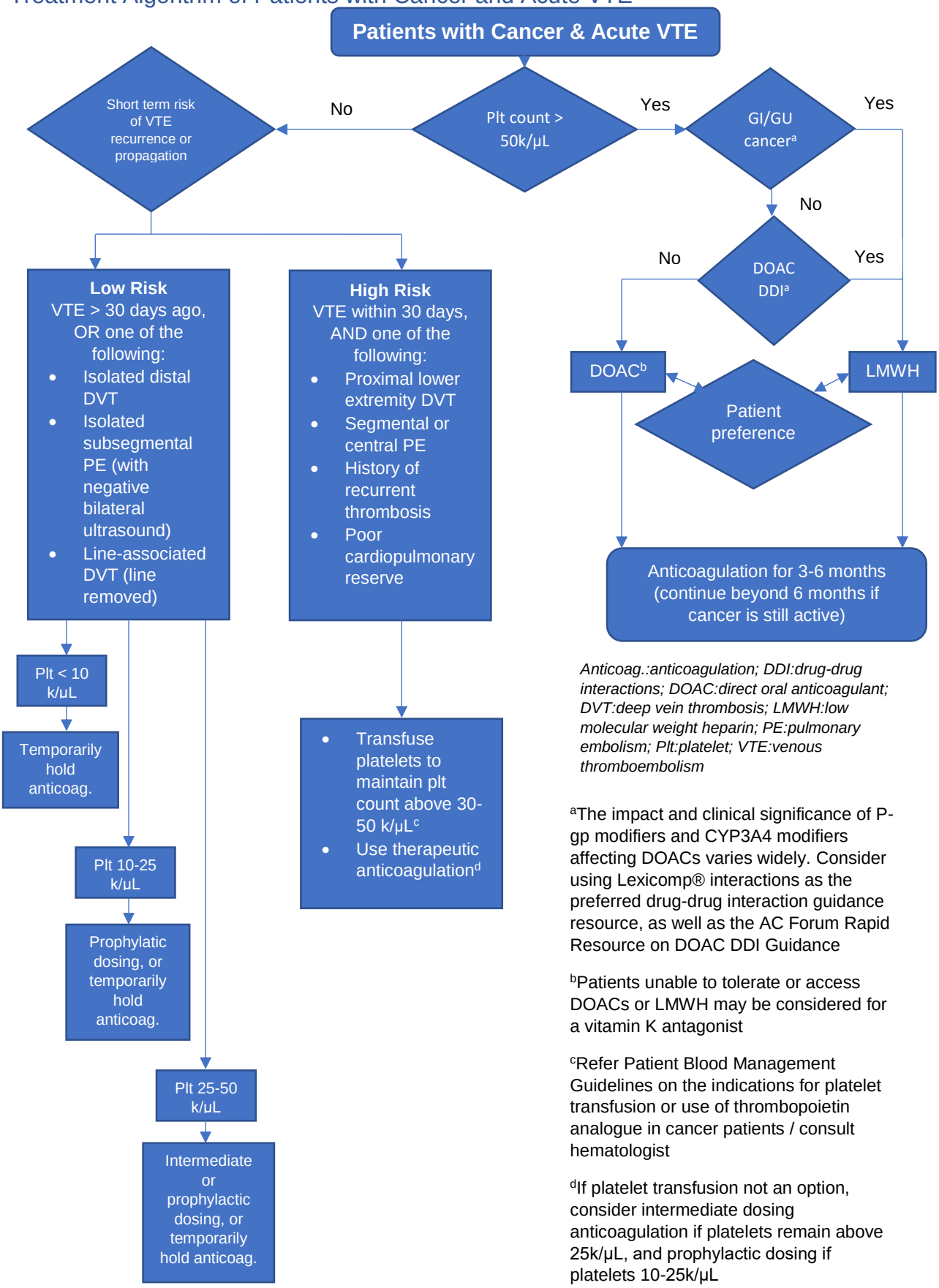
aPTT:activated partial thromboplastin time; LMWH: low molecular weight heparin; UFH:unfractionated heparin; VTE: venous thromboembolism

†Women treated with LMWH may be switched to a continuous infusion of UFH at least 36 h before the induction of labour or caesarean section.

§Consideration should be given to the use of intravenous UFH when VTE occurs at term which is more easily manipulated[17]. Based on recommendations for anticoagulant management around the time of delivery (prevention/treatment) of various guidelines, switching to UFH at 37 weeks may be considered to allow for more options with respect to labour analgesia. Conversion to UFH in anticipation of delivery can minimize the time off anticoagulants[18].

Anticoagulants in Cancer Patients[11]

Treatment Algorithm of Patients with Cancer and Acute VTE[19]



Anticoag.:anticoagulation; DDI:drug-drug interactions; DOAC:direct oral anticoagulant; DVT:deep vein thrombosis; LMWH:low molecular weight heparin; PE:pulmonary embolism; Plt:platelet; VTE:venous thromboembolism

[a]The impact and clinical significance of P-gp modifiers and CYP3A4 modifiers affecting DOACs varies widely. Consider using Lexicomp® interactions as the preferred drug-drug interaction guidance resource, as well as the AC Forum Rapid Resource on DOAC DDI Guidance

[b]Patients unable to tolerate or access DOACs or LMWH may be considered for a vitamin K antagonist

[c]Refer Patient Blood Management Guidelines on the indications for platelet transfusion or use of thrombopoietin analogue in cancer patients / consult hematologist

[d]If platelet transfusion not an option, consider intermediate dosing anticoagulation if platelets remain above 25k/μL, and prophylactic dosing if platelets 10-25k/μL

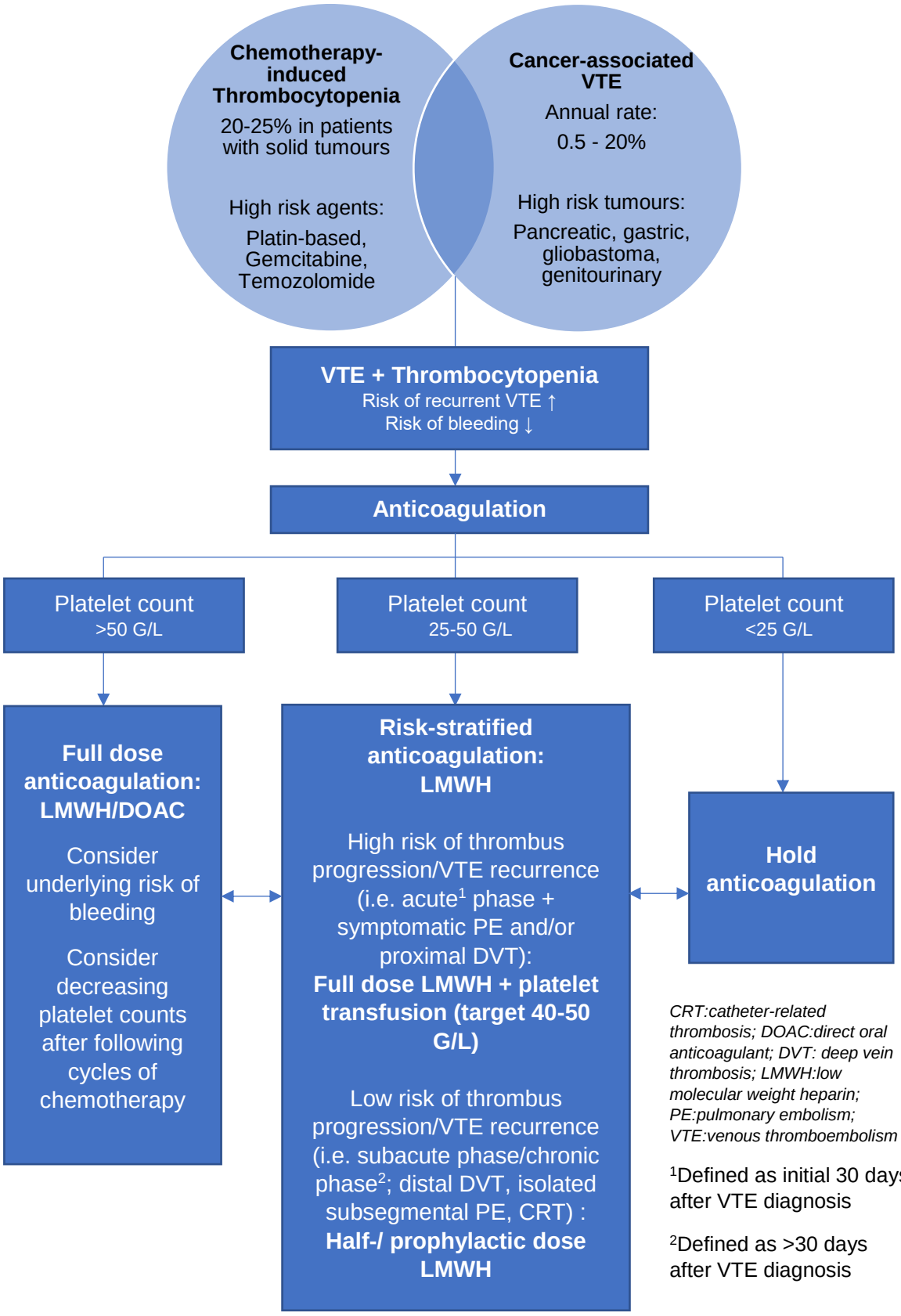
Guideline Recommendations[20]			
	2020 National Comprehensive Cancer Network (NCCN)	**2020 American Society of Clinical Oncology (ASCO)***	**2018 International Society on Thrombosis and Haemostasis (ISTH)***
Acute VTE Treatment	DOAC (edoxaban, apixaban and rivaroxaban) preferred for patients without gastric or gastroesophageal lesions	Initial anticoagulation (first 5-10 days); LMWH or rivaroxaban preferred Long-term (at least 6 months): LMWH, edoxaban or rivaroxaban preferred	DOACs (edoxaban and rivaroxaban) suggested for low risk of bleeding and no drug-drug interactions, LMWH suggested otherwise
GI/GU cancer	LMWH preferred for patients with gastric or gastroesophageal lesions	There is an increase in major bleeding risk with DOACs, particularly observed in GI and potentially GU malignancies	LMWH suggested in patients with luminal GI cancers with an intact primary, patients at risk of bleeding from the GU tract, bladder or nephrostomy tubes or patients with active GI mucosal abnormalities

DOAC:direct oral anticoagulant; GI:gastrointestinal; GU:genitourinary; LMWH:low molecular weight heparin; VTE:venous thromboembolism

*The recommendations from ASCO and ISTH were made before apixaban was compared to LMWH

Anticoagulation in Chemotherapy-induced Thrombocytopenia[21]

Treatment algorithm for cancer-associated VTE in patients with chemotherapy-induced thrombocytopenia (based on guidance from the Standardized Scientific Committee of the International Society on Thrombosis and Haemostasis)



CRT:catheter-related thrombosis; DOAC:direct oral anticoagulant; DVT: deep vein thrombosis; LMWH:low molecular weight heparin; PE:pulmonary embolism; VTE:venous thromboembolism

[1]Defined as initial 30 days after VTE diagnosis

[2]Defined as >30 days after VTE diagnosis

Anticoagulants in Left Ventricular Thrombus (LVT)[22]

Guideline Recommendations

LVT Setting	Recommendation	Class of Recommendation/ Level of Evidence
2012 CHEST Guidelines[1]		
LVT or high risk for LVT[b] + anterior MI with NO STENT	Warfarin (INR 2-3) + low dose aspirin x 3 months[c] Then discontinue warfarin and continue DAPT for up to 12 months per ACS recommendations	Grade 1B
LVT or high risk for LVT[b] + anterior MI with BMS	Warfarin (INR 2-3) + low dose aspirin + clopidogrel x1 month[a]; Then warfarin + SAPT for 2nd and 3rd month; Then discontinue warfarin and continue DAPT for up to 12 months per ACS recommendations	Grade 2C
LVT or high risk for LVT[b] + anterior MI with DES	Warfarin (INR 2-3) + low dose aspirin + clopidogrel x3-6 months[a]; Then discontinue warfarin and continue DAPT for up to 12 months per ACS recommendations	Grade 2C
LVT + systolic LV dysfunction without established CAD	Warfarin (INR 2-3) for at least 3 months[c]	Grade 2C
High risk for LVT[b] without established CAD	No antiplatelet therapy or warfarin	Grade 2C
2013 ACC/AHA STEMI Guidelines[3]		
Asymptomatic LV mural thrombus or at high risk for LVT[b] + STEMI	Warfarin (INR 2-3, or 2-2.5 in triple therapy) x3 months[c]	Class IIa
2014 AHA/American Stroke Association Guidelines[2]		
Ischaemic stroke or TIA in setting of acute MI complicated by LV mural thrombus	Warfarin (INR 2-3) x3 months[c]	Class I
Ischemic stroke or TIA in setting of acute MI complicated by LV mural thrombus INTOLERANT to VKA therapy	LMWH, dabigatran, rivaroxaban, or apixaban x3 months[c]	Class IIb
Ischemic stroke or TIA in setting of acute MI with high risk for LVT[b]	May consider warfarin (INR 2-3) x3 months[c]	Class IIb

Ischemic stroke or TIA in sinus rhythm who have left atrial or LVT	Warfarin (INR 2-3) x3 months[c]	Class I
Ischaemic stroke or TIA in sinus rhythm at high risk for LVT[b]	Effectiveness of anticoagulation is uncertain; treatment choice should be individualized	Class IIb
2017 European Society of Cardiology STEMI Guidelines[4]		
LVT	Anticoagulation[d] for up to 6 months guided by repeated imaging	Class IIa

Treatment of LVT with DOACs in the Literature[e]

Publication Date/Type	No. of patients	Anticoagulant	Treatment Outcomes
Feb 2018; Systematic Review[5]	15	Rivaroxaban	73% resolved; Duration ranged 7-436 days
Jan 2019; Systematic Review[6]	41	Apixaban;11 pts Rivaroxaban; 21 pts Dabigatran; 9 pts	Apixaban: 100% resolved; median 36 days Rivaroxaban: 81% resolved; median 40 days Dabigatran: 89% resolved; median 24 days
April 2019; Systematic Review[7]	52	Apixaban; 10 pts Rivaroxaban; 30 pts Dabigatran; 12 pts	Apixaban: 100% resolved; median 36 days Rivaroxaban: 90% resolved; median 90 days Dabigatran: 75% resolved; median 28 days
April 2019; Retrospective cohort study[8]	108	Apixaban; 3 pts Rivaroxaban; 1 pt Warfarin; 94 pts LMWH; 10 pts	Apixaban: 100% resolved at 1yr. Rivaroxaban: 100% resolved at 1yr. Warfarin: 75% resolved at 1yr. LMWH: 40% resolved at 1yr. Duration varied
May 2019; Retrospective cohort study[9]	35	Apixaban; 16 pts Rivaroxaban; 17 pts Dabigatran; 2 pts	Apixaban: 76% resolved Rivaroxaban: 76% resolved Dabigatran: 50% resolved; Duration and follow-up varied
April 2020; Retrospective cohort study[10] (RED VELVT)	514	DOAC; 121 pts Warfarin; 236 pts (64 pts switched between treatment groups, 93 pts not treated)	DOAC: 17 SSE events, 14 deaths, 8 bleeds Warfarin: 14 SSE events, 32 deaths, 19 bleeds; Duration and follow-up varied

ACS:acute coronary syndrome; BMS:bare metal stent; CAD:coronary artery disease; DAPT:dual antiplatelet therapy; DES:drug eluting stent; DOAC:direct oral anticoagulant; INR:international normalized ratio; LMWH:low molecular weight heparin; MI:myocardial infarction; RCT:randomized controlled trials; SAPT:single antiplatelet therapy; STEMI:ST elevated MI; TIA:transient ischemic attack; VKA:vitamin K antagonist

[a] Must weigh potential risks vs. benefits of adding warfarin to DAPT. Duration of triple therapy, if chosen, should be minimized. For pts with large anterior STEMI and demonstrated LV thrombus, addition of warfarin to DAPT would be expected to result in 44 fewer nonfatal strokes and 15 more nonfatal extracranial bleeds per 1,000 patients treated. For pts with large anterior STEMI and no LV thrombus, adding warfarin to DAPT would prevent 7 nonfatal strokes at a cost of 15 nonfatal extracranial bleeds per 1,000 patients treated.

[b] EF <40%, antero-apical wall motion abnormality.

[c] Patients with persistent mobile or protruding thrombus visualized by echo or another imaging modality may remain at increased risk for stroke and other embolic events beyond 3 months.

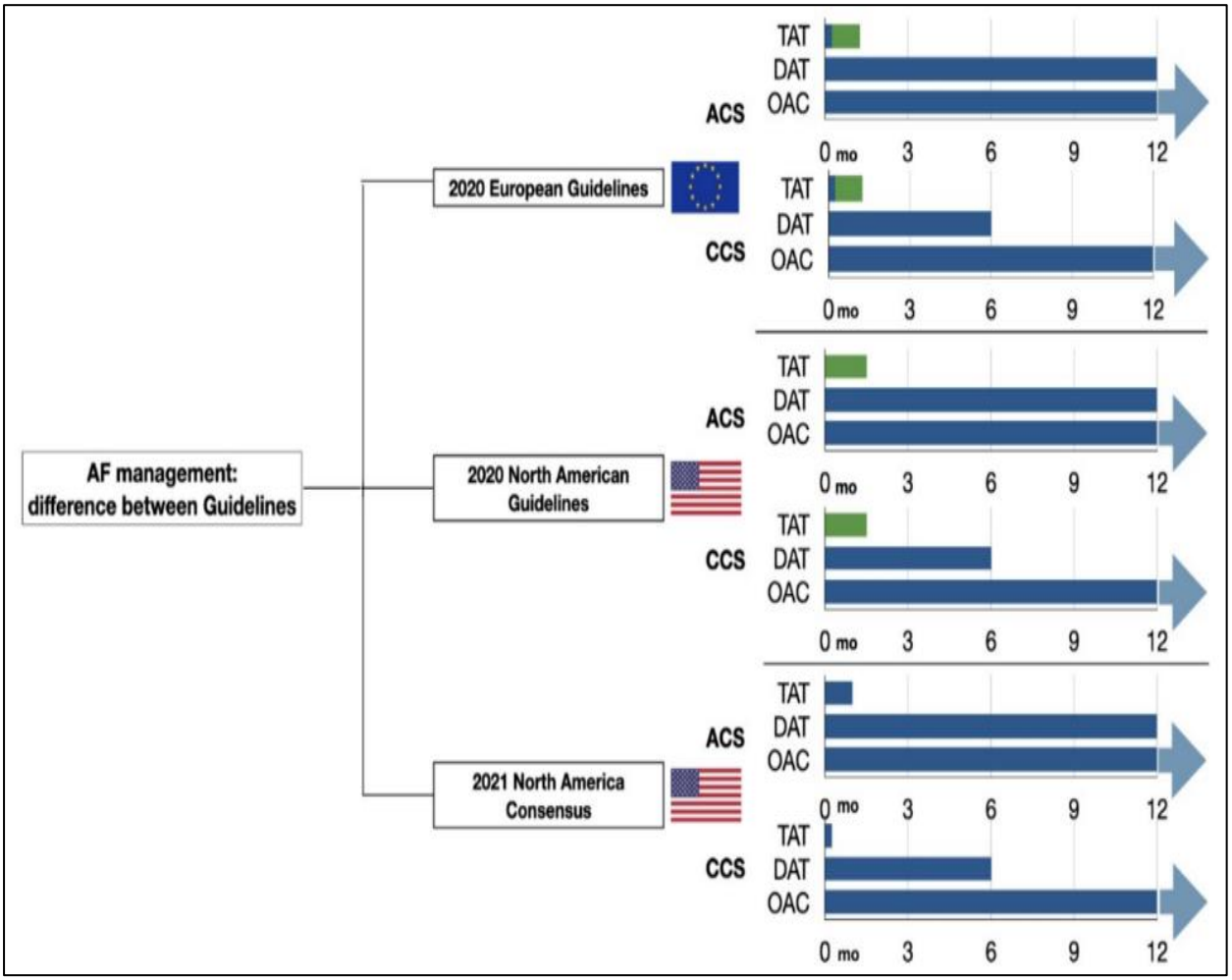
[d] No recommendation on type of OAC. Bleeding risk and need for concomitant antiplatelet therapy should be considered.

[e] Referenced literature employed various dosing strategies. (e.g. standard vs. initial VTE treatment DOAC dosing or warfarin with or without heparin overlap). Specific dosing recommendations are outside the scope of this review and have been omitted from this document. However, factors that should be considered when selecting a strategy include whether thrombus is mobile/protruding, appears “new” in nature, and patient bleeding risks.

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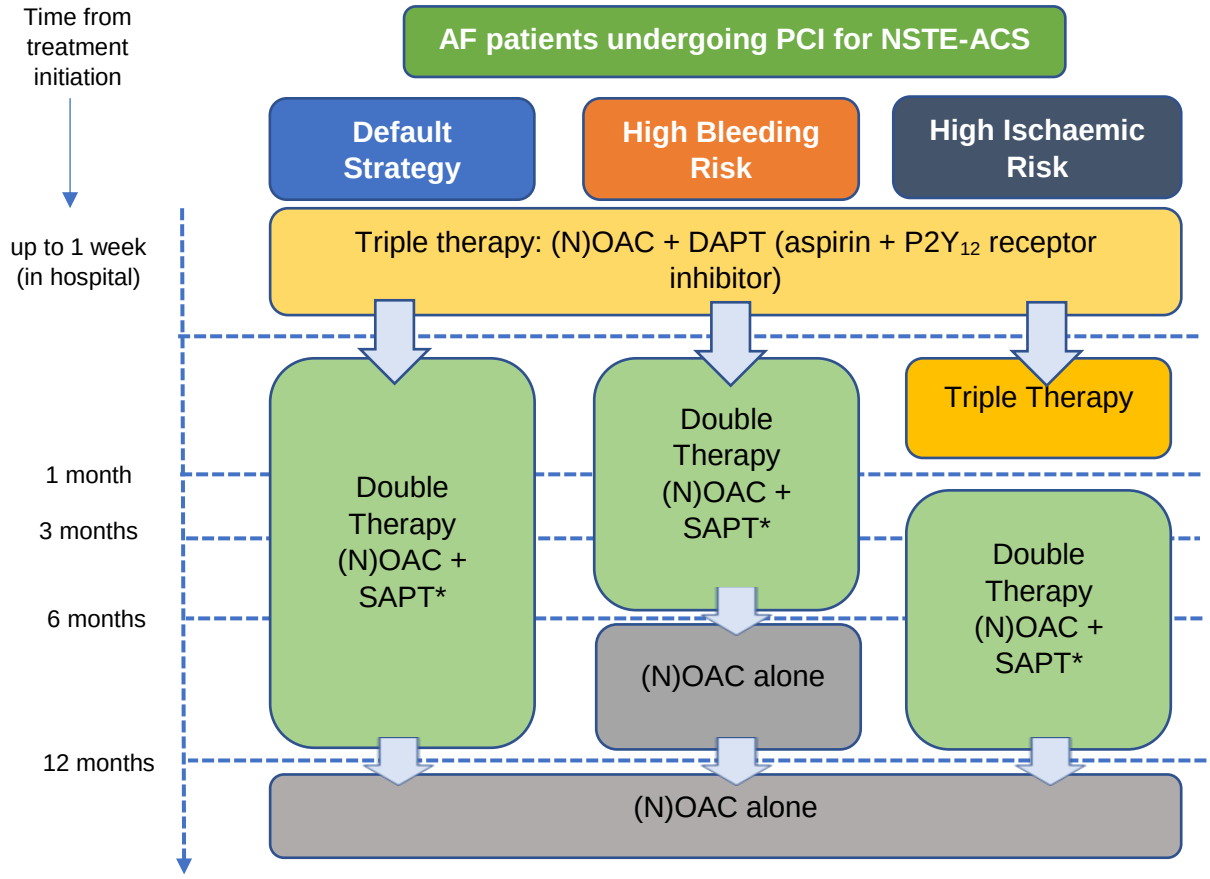
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Anticoagulants in AF Patients with Acute Coronary Syndrome (ACS)[23]



ACS:acute coronary syndrome; AF:atrial fibrillation; CCS:chronic coronary syndrome; DAT:dual-antithrombotic therapy; OAC:oral anticoagulant; TAT:triple-antithrombotic therapy.

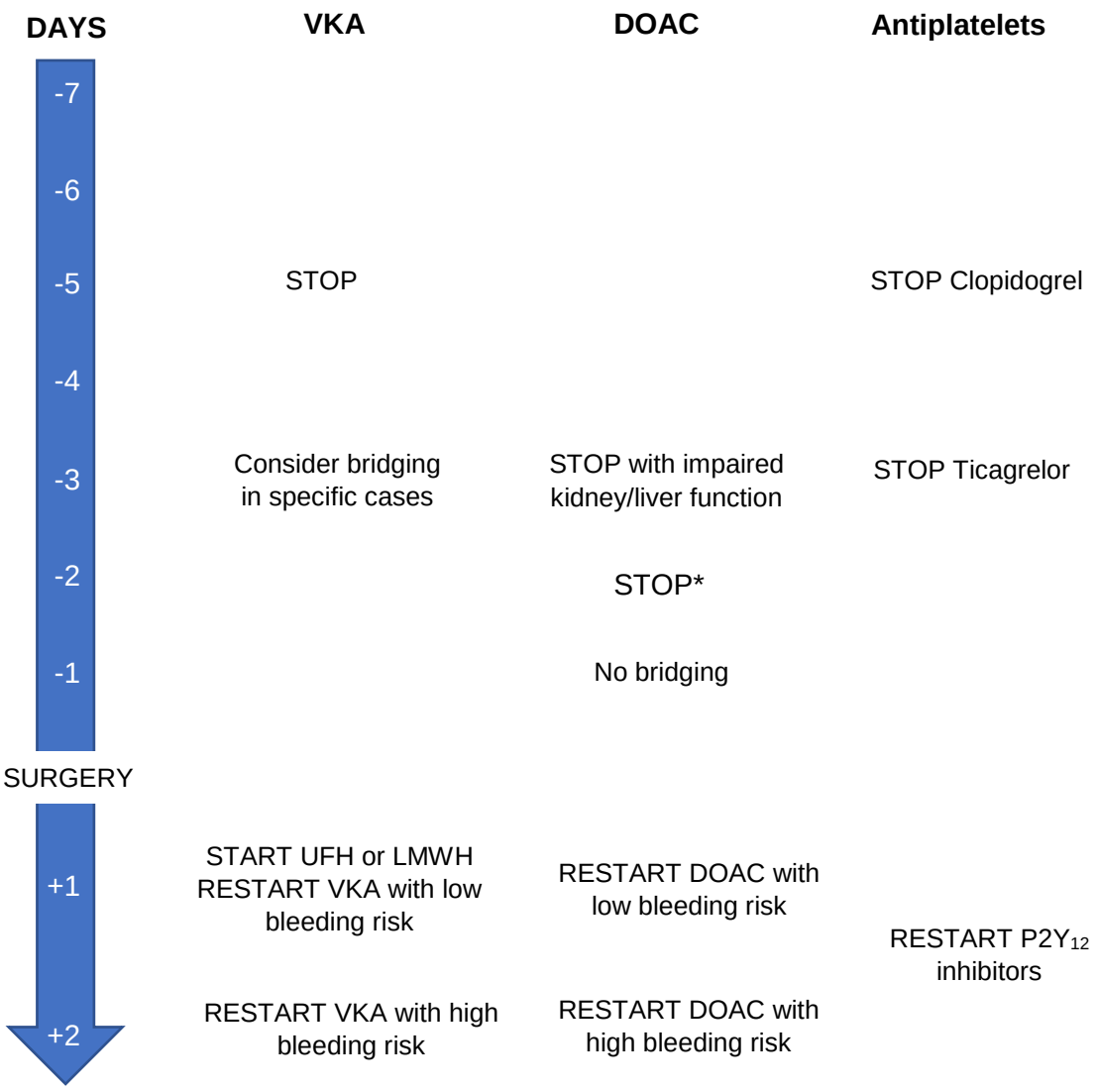
Anticoagulants in AF Patients [Non ST-Elevation Myocardial Infarction(NSTEMI)][24]



AF:atrial fibrillation; DAPT:dual antiplatelet therapy; NOAC:non-vitamin K antagonist oral anticoagulant; OAC:oral anticoagulant; SAPT:single antiplatelet therapy

*Clopidogrel preferred

Pre-operative Management of Anticoagulants[25]



DOAC:direct oral anticoagulant; LMWH:low molecular weight heparin; UFH:unfractionated heparin; VKA:vitamin K antagonist

*May be continued in patients undergoing AF ablation

Anticoagulants during Dental Procedures[26]

Dental procedures that are **unlikely to cause bleeding**	Dental procedures that are **likely to cause bleeding**	
	Low risk of post-operative bleeding complications	**Higher risk** of post-operative bleeding complications
Local anaesthesia by infiltration, intraligamentary or mental nerve block[a] Local anaesthesia by inferior dental block or other regional nerve blocks[a,b] Basic periodontal examination (BPE)[c] Supragingival removal of plaque, calculus and stain Direct or indirect restorations with supragingival margins Endodontics – orthograde Impressions and other prosthetics procedures Fitting and adjustment of orthodontic appliances	Simple extractions (1-3 teeth, with restricted wound size)[d] Incision and drainage of intraoral swellings Detailed six point full periodontal examination Root surface instrumentation (RSI) and subgingival scaling Direct or indirect restorations with subgingival margins	Complex extractions[e], adjacent extractions that will cause a large wound or more than 3 extractions at once Flap raising procedures : • Elective surgical extractions • Periodontal surgery • Preprosthetic surgery • Periradicular surgery • Crown lengthening • Dental implant surgery Gingival recontouring Biopsies

[a] Local anaesthesia should be delivered using an aspirating syringe and should include a vasoconstrictor, unless contraindicated. Note that other methods of local anaesthetic delivery are preferred over regional nerve blocks, whether the patient is taking an anticoagulant or not.

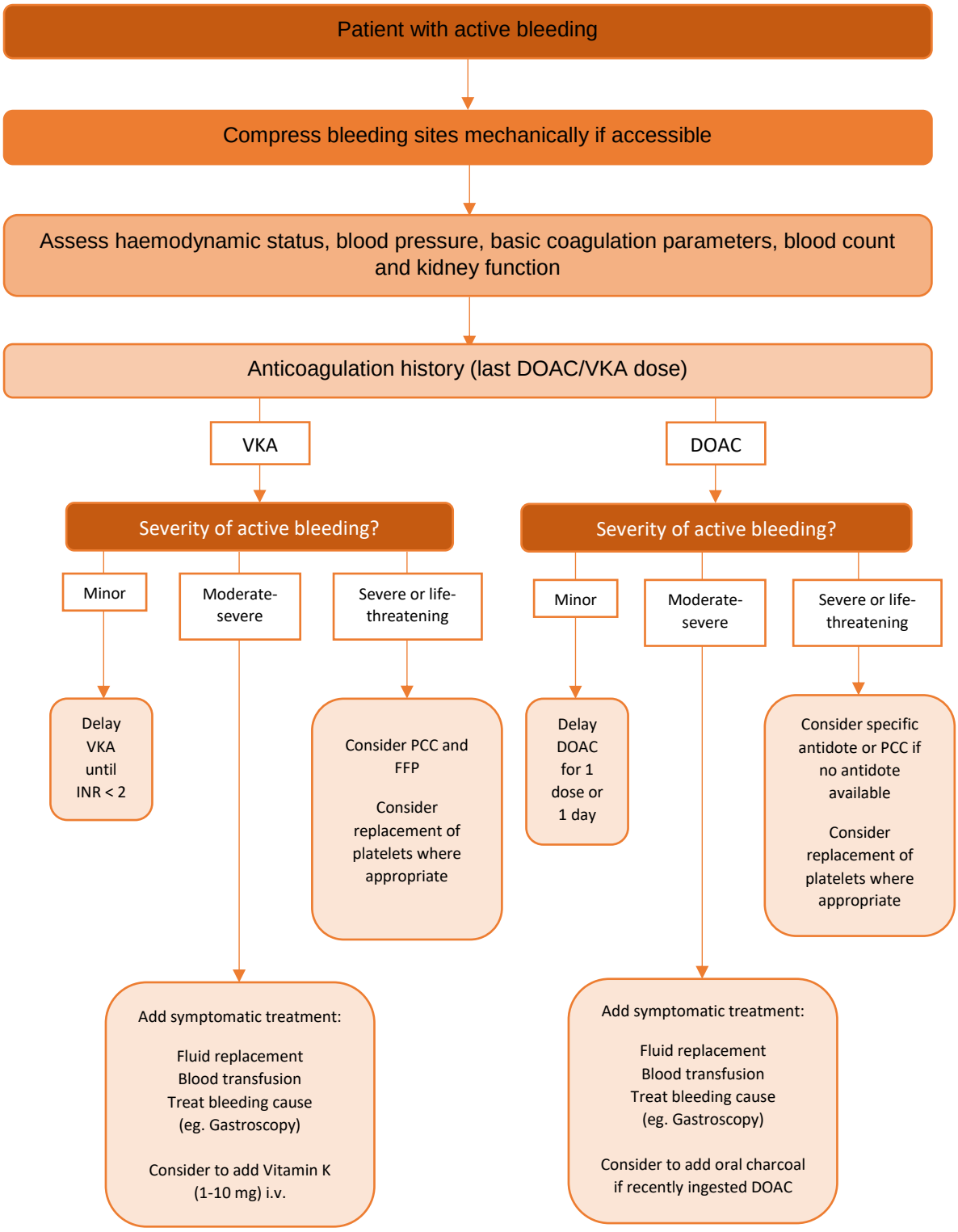
[b] There is no evidence to suggest that an inferior dental block performed on an anticoagulated patient poses a significant risk of bleeding. However, for patients taking warfarin, if there are any indications that the patient has an unstable INR, or other signs of excessive anticoagulation, an INR should be requested before the procedure.

[c] Although a BPE can result in some bleeding from the gingival margins, this is extremely unlikely to lead to complications.

[d] Simple extractions refers to those which are expected to be straightforward, without surgical complications.

[e] Complex extractions refers to those which may be likely to have surgical complications.

Management of Bleeding[27]



DOAC:direct oral anticoagulant; FFP:fresh frozen plasma; i.v:intravenous; PCC:prothrombin complex concentrates; VKA:vitamin K antagonist

Notes:

- 3-FPCC can be used if 4–FPCC is not available
- 3-FPCC or 4–FPCC can be used if Idarucizumab is not available

Agents to Stop Bleeding[28]

Agent	Dose	Comments
Vitamin K	1 – 10 mg IV/PO	• Infusion reactions rare; administer over 20-30 min • Takes 6 (IV) to 24 (PO) hours to reverse warfarin • Large doses can cause warfarin resistance on resumption • Subcutaneous or intramuscular administration not recommended
Protamine sulfate	12.5 – 50 mg IV	• Full reversal of unfractionated heparin • 60-80% reversal of LMWH • No reversal of fondaparinux
Platelets	1 apheresis unit 5-8 whole blood units	• Used in patients receiving antiplatelet therapy • Raise platelet count by 30 x 10^9/L • Goal platelet count 50 – 100 x 10^9/L (indication dependent)
Frozen plasma (FFP)	10 – 30 mL/kg (1 unit = ~250 ml)	• Replaces all coagulation factors, but cannot fully correct ○ Hemostasis usually requires coagulation factor levels ~30% ○ Factor IX may only reach 20% • Short half-life, may need repeat dosing after 6 hours • Large volume, can take hours to thaw and infuse
Prothrombin complex concentrates (PCC)	25 – 50 units/kg IV (lower doses studied)	• Rapid, complete INR correction in warfarin-treated patients • Small volume infusion over 10-30 minutes • Risk of thrombosis 1.4%; contraindicated with history of HIT • Short half-life, may need repeat dose after 6 hours • Consider concurrent FFP If 3-factor PCC used
Recombinant factor VIIa (rFVIIa)	15-90 micrograms/kg (lower doses studied)	• Rapid infusion of small volume • Rapid complete INR correction in warfarin patients, but may not correct bleeding because only restores FVIIa • Risk of thrombosis 5-10% • Short half-life, may need repeat dose after 2 hours
Aminocaproic acid	4-5 g IV/po over 1 hr, then 1g/hr x 8 hrs (maximum dose 30 g/24 hrs)	• May increase risk of thrombosis • May accumulate in patients with renal impairment, reduce loading dose or infusion rate • Caution or avoid use in hematuria due to upper urinary tract origin
Tranexamic acid	1300 mg po q8h	• Labeled indication for menorrhagia • Use for other indications currently off label

INR:international normalized ratio; IV:intravenous; LMWH:low molecular weight heparin; PO:oral

Reversal Strategy for Warfarin[29]

Management of patients on warfarin therapy with high INR and without bleeding

INR	Bleeding risk	Warfarin	Vitamin K	PTX-VF	Check INR	Comments
INR higher than therapeutic range but < 4.5		Reduce or omit next dose				Resume warfarin at reduced dose when INR reaches therapeutic range
INR 4.5 – 10.0	Low	Cease			Within 24h	
	High	Cease	1 – 2 mg PO or 0.5 -1 mg IV*			
INR > 10.0	Low	Cease	3 – 5 mg PO or IV **		Within 12h	
	High	Cease	3 – 5 mg PO or IV**	15 – 30 IU/kg		

h:hours; INR:international normalized ratio

*< 12 years of age give 0.03 mg/kg IV (max 1 mg)

** < 12 years of age give 0.3 mg/kg IV (max 10 mg)

Suggested doses of Prothrombinex-VF to reverse the anticoagulant effect of warfarin according to initial and target INR

Patient's initial INR	1.5 – 2.5	2.6 – 3.5	3.6 – 10.0	>10.0
Target INR 0.9 – 1.3	30 IU/kg	35 IU/kg	50 IU/kg	50 IU/kg
Target INR 1.4 – 2.0	15 IU/kg	25 IU/kg	30 IU/kg	40 IU/kg

INR:international normalized ratio

Management of patients on warfarin therapy WITH bleeding

INR	Bleeding risk	Warfarin	Vitamin K	PTX-VF	FFP	Check INR	Comments
INR ≥ 1.5 with life-threatening (critical organ) bleeding		Cease	5 – 10 mg IV**	50 IU/kg	150 – 300 mL If PTX-VF not available administer FFP 15 mL/kg	In 20 mins	Resume warfarin when bleeding ceased and adjust dose to maintain INR within therapeutic range
INR ≥ 2.0 with clinically significant bleeding (not life-threatening)		Cease	5 – 10 mg IV**	35 - 50 IU/kg	If PTX-VF not available administer FFP 15 mL/kg	In 20 mins	
Any INR with minor bleeding or INR > 4.5 with minor bleeding	Low	Cease				In 24h	Resume warfarin at reduced dose when INR reaches the therapeutic range
	High	Cease	Consider 1 – 2 mg PO or 0.5 – 1 mg IV*				

h:hours; INR:international normalized ratio; IV:intravenous; PO:oral; mins:minutes;

*< 12 years of age give 0.03 mg/kg IV (max 1 mg)

** < 12 years of age give 0.3 mg/kg IV (max 10 mg)

Reversal Strategy for DOAC[11]

Dabigatran

- Idarucizumab : 5 g IV x 1
- Consider activated charcoal for known recent ingestions within 2 – 4 hours
- If idarucizumab not available : aPCC or PCC 50 units/kg IV x 1 (max 4,000 units)
- Hemodialysis (if haemodynamically stable and idarucizumab not available)

Factor Xa Inhibitor

Andexanet alfa dosed per below :

Drug	Last Dose	Time from last dose	
		< 8 hrs or unknown	≥ 8 hrs
Rivaroxaban	≤10 mg	Low Dose	Low Dose
	>10 mg or unknown	High Dose	
Apixaban	≤5 mg	Low Dose	Low Dose
	>5 mg or unknown	High Dose	
Edoxaban[a]	Any Dose	High Dose	Limited data

Consider activated charcoal for known recent ingestions within 2-4 hours

If andexanet alfa not available :
4F-PCC 25-50 units/kg (ICH) or 2000 units IV x 1

Low Dose – 400 mg IV bolus at target rate of 30 mg/min followed by continuous infusion at 4 mg/min for up to 120 minutes.

High Dose – 800 mg IV bolus at target rate of 30 mg/min followed by continuous infusion at 8 mg/min for up to 120 minutes

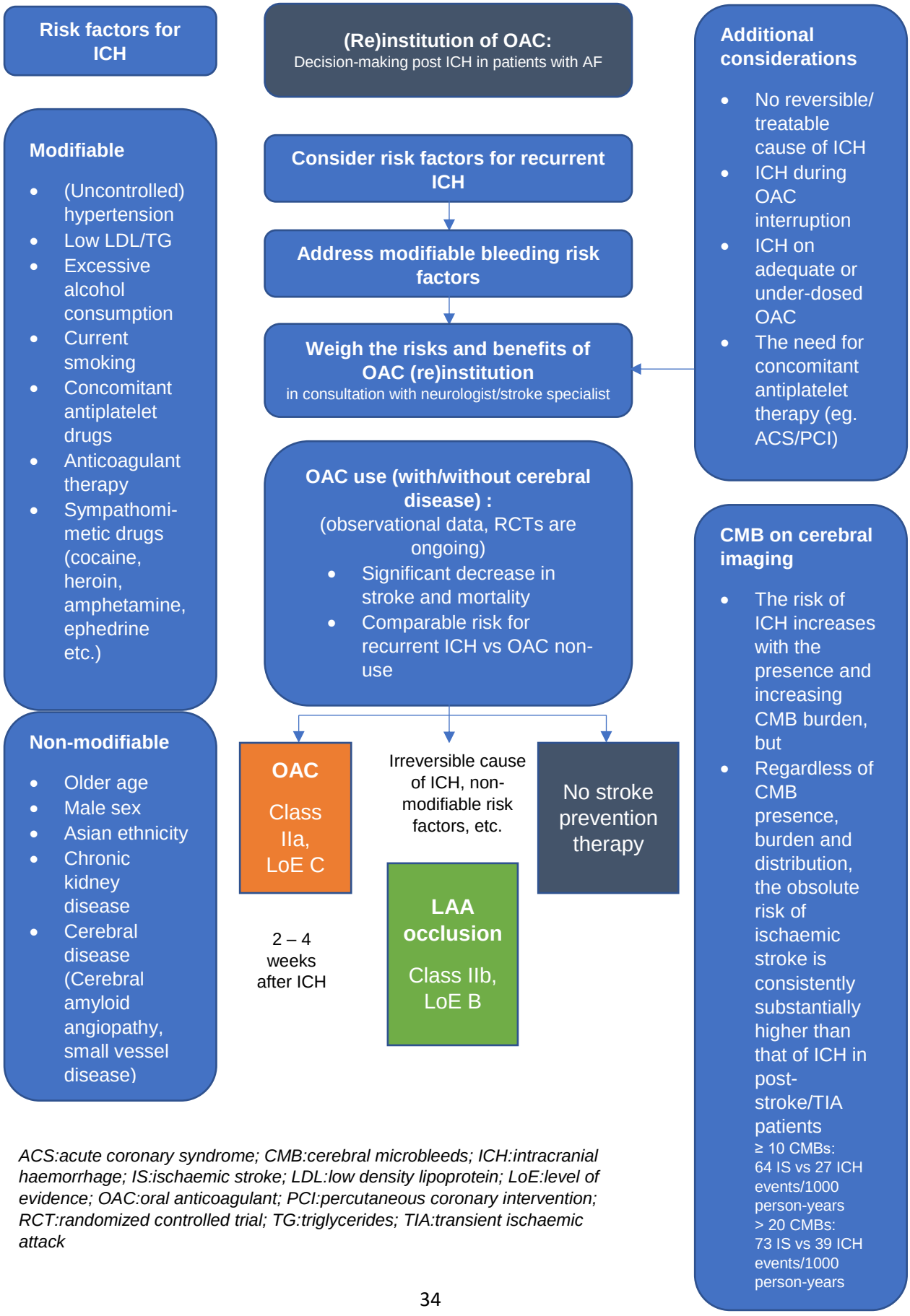
[a]Andexanet alfa is only approved by the U.S. Food and Drug Administration for apixaban and rivaroxaban-induced bleeding. Off-label use of andexanet alfa for reversal of edoxaban may be considered.

Considerations

Andexanet alfa: Clinical judgement advised when selecting dose regimens based on time cut-offs in patients presenting with acute or chronic renal dysfunction and life-threatening bleeding, receiving non-renally dose adjusted baseline DOAC regimens.

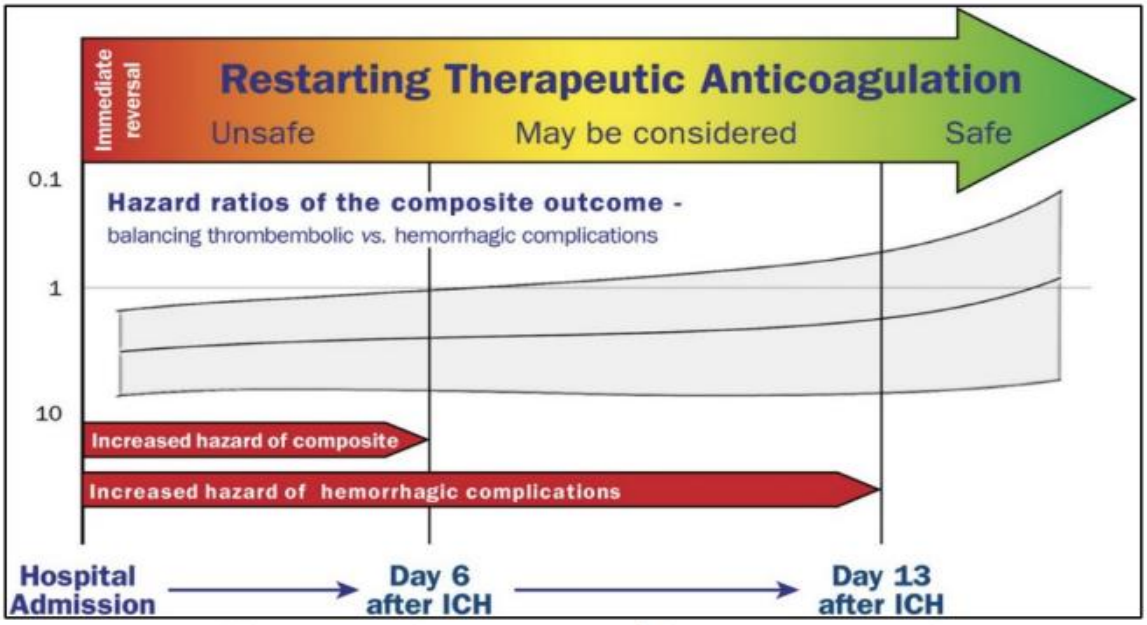
Re-Initiation of Anticoagulants Post ICH

Non-Heart Valve Patients[27]



ACS:acute coronary syndrome; CMB:cerebral microbleeds; ICH:intracranial haemorrhage; IS:ischaemic stroke; LDL:low density lipoprotein; LoE:level of evidence; OAC:oral anticoagulant; PCI:percutaneous coronary intervention; RCT:randomized controlled trial; TG:triglycerides; TIA:transient ischaemic attack

Patients with Mechanical Heart Valve[30]



Appendix

Table 3: CHA_2DS_2-VASc score[31]

(a) Risk factors for stroke and thromboembolism in non-valvular AF	
'Major' risk factors	**'Clinically relevant non-major' risk factors**
Previous stroke, TIA, or systemic embolism Age ≥ 75 years	Heart failure or moderate to severe LV systolic dysfunction (e.g. LV EF ≤ 40%) Hypertension – Diabetes mellitus Female sex – Age 65 – 74 years Vascular disease[a]

(b) Risk factor-based approach expressed as a point-based scoring system, with the acronym CHA_2DS_2-VASc (Note: maximum score is 9 since age may contribute 0,1 or 2 points)	
Risk factor	**Score**
Congestive heart failure/LV dysfunction	1
Hypertension	1
Age ≥ 75	2
Diabetes mellitus	1
Stroke/TIA/thromboembolism	2
Vascular disease	1
Age 65 – 74	1
Sex category (i.e. female sex)	1
Maximum score	**9**

(c) Adjusted stroke rate according to CHA_2DS_2-VASc		
CHA_2DS_2-VASc	**Patients (n=7329)**	**Adjusted stroke rate (%/year)[b]**
0	1	0%
1	422	1.3%
2	1230	2.2%
3	1730	3.2%
4	1718	4.0%
5	1159	6.7%
6	679	9.8%
7	294	9.6%
8	82	6.7%
9	14	15.2%

AF:atrial fibrillation; EF:ejection fraction (as documented by echocardiography, radionuclide ventriculography, cardiac catheterization, cardiac magnetic resonance imaging, etc.); LV:left ventricular; TIA:transient ischaemic attack

[a]Prior myocardial infarction, peripheral artery disease, aortic plaque. Actual rates of stroke in contemporary cohorts may vary from these estimates.

[b]Based on Lip et. al.[32]

Table 4: HASBLED Score[31,33]

Letter	Clinical characteristic[a]	Points awarded
H	Hypertension	1
A	Abnormal renal and liver function (1 point each)	1 or 2
S	Stroke	1
B	Bleeding	1
L	Labile INRs	1
E	Elderly (eg. age > 65 years)	1
D	Drugs or alcohol (1 point each)	1 or 2
		Maximum 9 points

INR: international normalized ratio

[a]'Hypertension' is defined as systolic blood pressure > 160 mmHg. 'Abnormal renal function' is defined as the presence of chronic dialysis or renal transplantation or serum creatinine ≥ 200 µmol/L. 'Abnormal liver function' is defined as chronic hepatic disease (eg. cirrhosis) or biochemical evidence of significant hepatic derangement (eg. bilirubin >2x upper limit of normal, in association with aspartate aminotransferase/alanine aminotransferase/alkaline phosphatase > 3x upper limit normal, etc.). 'Bleeding' refers to previous bleeding history and/or predisposition to bleeding, eg. bleeding diathesis, anaemia, etc. 'Labile INRs' refers to unstable/high INRs or poor time in therapeutic range (eg. < 60%). Drugs/alcohol use refers to concomitant use of drugs, such as antiplatelet agents, non-steroidal anti-inflammatory drugs, or alcohol abuse, etc.

Table 5: Wells' Criteria for Deep Vein Thrombosis (DVT)[34]

Clinical Characteristic	Score
Active cancer (patient receiving treatment for cancer within the previous 6 months or currently receiving palliative treatment)	1
Paralysis, paresis or recent plaster immobilization of the lower extremities	1
Recently bedridden for 3 days or more, or major surgery within the previous 12 weeks requiring general or regional anesthesia	1
Localized tenderness along the distribution of the deep venous system	1
Entire leg swollen	1
Calf swelling at least 3 cm larger than that on the asymptomatic side (measured 10 cm below tibial tuberosity)	1
Pitting edema confined to the symptomatic leg	1
Collateral superficial veins (nonvaricose)	1
Previously documented deep-vein thrombosis	1
Alternative diagnosis at least as likely as deep-vein thrombosis	-2

*A score of two or higher indicates that the probability of deep-vein thrombosis is likely; a score of less than two indicates that the probability of deep-vein thrombosis is unlikely. In patients with symptoms in both legs, the more symptomatic leg is used.

Table 6: Wells' Criteria for Pulmonary Embolism (PE)[35–37]

Clinical Characteristic	Score
Active cancer (treatment within 6 months or palliative)	1
Surgery or bedridden for ≥ 3 days during past 4 weeks	1.5
History of deep venous thrombosis or pulmonary embolism	1.5
Hemoptysis	1
Heart rate of > 100 beats/min	1.5
Pulmonary embolism judged to be the most likely diagnosis	3
Clinical signs and symptoms compatible with deep vein thrombosis	3

*A total score of ≤ 4 indicates a low probability of pulmonary embolism (also termed "pulmonary embolism unlikely" in other reports[35,38]. A score of 4.5 to 6 indicates a moderate probability of pulmonary embolism; a score of > 6 indicates a high probability of pulmonary embolism[35]

Table 7: Predictive Model for Chemotherapy-Associated Venous Thromboembolism (VTE) in the Ambulatory Setting (Khorana Score) [39,40]

Patient Characteristic	Points
Site of cancer	
Very high risk (Stomach, pancreas, primary brain tumour)	2
High risk (lung, lymphoma, gynecologic, bladder, testicular, renal tumors)	1
Prechemotherapy platelet count ≥ 350,000/μL	1
Hemoglobin level < 10 g/dL or use of red-cell growth factors	1
Prechemotherapy leukocyte count > 11,000/μL	1
Body mass index ≥ 35 kg/m^2	1

Calculate total score, adding points for each criterion in the model.
Interpretation: High risk ≥ 3 points; Intermediate risk, 1 to 2 points; Low risk, 0 points

Table 8: DASH Prediction Score for Recurrent VTE[41,42]

Letter	Clinical characteristic[a]	Points awarded
D	D-dimer abnormal (measured ~1 month after stopping anticoagulation)	No [0 point] Yes [2 points]
A	Age	≥ 50 years [0 point] ≤ 50 years [1 point]
S	Sex (Male patient)	No [0 point] Yes [1 point]
H	Hormone use at VTE onset (if female)	No [0 point] Yes [-2 point]

In a patient with previously diagnosed VTE who has completed a 3 - 6-month course of anticoagulation:
DASH ≤1:
Consider discontinuing anticoagulation, as this group has an annual recurrence risk of 3.1%.
DASH ≥2:
Consider continuing anticoagulation, as this group has an annual recurrence risk of 9.3%.

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